

Highlights

M3-Score: Fidelity, Memorization and Coverage as Separate Axes for Evaluating Generative Radiology Image Models

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- M3 reports fidelity, memorization and coverage as three separate axes, not one score.
- Each axis is read from a transformer block fixed a priori, never tuned on test outcomes.
- A DDPM scores well on fidelity while covering only 38% of the real manifold.
- k -NN manifold recall rises as generated diversity is removed, at all twelve blocks.
- Across a twentyfold change in evaluation size M3 drifts 1.05 $\times$ and FID 2.52 $\times$.

M3-Score: Fidelity, Memorization and Coverage as Separate Axes for Evaluating Generative Radiology Image Models

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ARTICLE INFO

Keywords:

generative models
medical image synthesis
evaluation metrics
maximum mean discrepancy
RadioDINO
diffusion models

ABSTRACT

Evaluating generative models for radiology image synthesis is harder than evaluating natural-image generators: the clinically relevant signal lies in rare, high-frequency pathological structure that natural-image embeddings under-represent, and a single scalar cannot distinguish a generator that renders plausible images from one that renders only a few kinds of plausible image. We present the **Medical Multi-axis Maximum Mean Discrepancy (M3-Score)**, an evaluation framework grounded in the frozen RadioDINO-s16 vision transformer (ViT-S/16) pre-trained on radiology images. M3 reports three separately defined quantities rather than one score, each read from a transformer block fixed *a priori* by representational depth and never tuned on test outcomes: *fidelity* (unbiased multi-bandwidth RBF MMD², final block), *memorization* (nearest-neighbor distances at 75% depth), and *coverage* (the fraction of real images with a generated neighbour inside their own *k*-NN radius, at 33% depth). Because slices from one patient are near-duplicates, every reference set here is drawn round-robin across subjects rather than by filename order; on a subject-diverse BraTS reference the three axes order five comparison sets of increasing severity exactly, and a held-out real probe correctly registers no shift (MMD² = 0.000, permutation $z = -0.6$). The axes are not redundant: an unconditional DDPM attains a modest fidelity distance (MMD² = 0.073, bootstrap 95% CI [0.071, 0.080]) while covering only 38% of the real manifold, the failure mode a fidelity score alone cannot express. We report a negative control that motivates the coverage estimator: the conventional *k*-NN manifold recall of Kynkäänniemi *et al.* rises as generated diversity is removed, at all twelve encoder blocks ($\rho = +0.70$ to $+1.00$), because its hyperspheres are fitted on the set being subsampled; the real-radius Coverage estimator of Naeem *et al.* falls monotonically at all twelve ($\rho = -1.00$). Three further results characterize the backbone and the estimator. Holding the scoring rule fixed and varying only the feature space, RadioDINO separates real brain MRI from in-domain DDPM output at ROC-AUC 0.819, where the InceptionV3 space underlying FID (0.555) and the CLIP space underlying CMMD (0.582) are near chance; the cross-modality and far-domain tiers are saturated for all three and rank nothing. On ordering, CMMD *rank-inverts* a cross-modality CT shift against a weak same-modality generator, while M3 and FID order both correctly. And across a twentyfold change in evaluation size on identical data, the M3 value drifts by 1.05 $\times$ while FID drifts by 2.52 $\times$, so FID values measured at different *N* are not comparable and M3 values are. Using real expert glioma masks with an area-matched, texture-controlled design, M3 is 1.4–1.8 $\times$ more responsive to tumor-region than to matched healthy-region perturbation where an Inception-MMD baseline is not ($\sim 1.05\times$). Three limitations bound the claims: FID is *more* repeatable than M3 at any fixed *N* < 500; the *a-priori* depths are fixed by representational position and we do not claim they are optimal; and the permutation test accompanying the fidelity axis tests a weak null that every generator rejects, so we report it descriptively rather than as a model-selection instrument. M3 is scoped to the radiology (CT/MRI) domain in which its backbone was trained.

1. Introduction

Generative models are increasingly used to address data scarcity of annotated medical imaging data, which continues to constrain clinical artificial intelligence (AI) development across modalities (Kebaili *et al.*, 2023; Kazerouni *et al.*, 2023). Strict patient privacy regulations, the high cost of expert annotation, severe class imbalance due to disease rarity, and logistical challenges in multi-center data sharing collectively limit the creation of large-scale, high-quality datasets (Finetti *et al.*, 2025; Akpinar *et al.*, 2024). After Generative Adversarial Networks (GANs) (Goodfellow *et al.*, 2014) established medical image synthesis as a practical tool, Denoising Diffusion Probabilistic Models (DDPMs) (Ho *et al.*, 2020) improved sample quality further, producing realistic and anatomically consistent images across MRI, CT,

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chest radiography, retinal fundus photography, dermoscopy, and ultrasound (Kazerouni et al., 2023; Khader et al., 2023; Pinaya et al., 2022). Downstream benefits, including improvements to segmentation, classification, and reconstruction performance through synthetic data augmentation, have been demonstrated across these modalities, particularly for rare-pathology cohorts (Finetti et al., 2025; Akpinar et al., 2024; Kebaili et al., 2023).

Yet, deploying generative models for medical image synthesis in clinical or research pipelines imposes a validation burden that conventional natural-image benchmarks do not meet. A synthetically generated medical image may attain excellent scores on standard quantitative metrics while still containing anatomically implausible structures, globally inconsistent intensity distributions, or clinically hazardous hallucinations (Kelkar et al., 2023; Deo et al., 2025). These artifacts are immediately evident to domain experts yet remain undetectable by conventional evaluation measures.

1.1. Limitations of Existing Evaluation Metrics

The Fréchet Inception Distance (FID) (Heusel et al., 2017) and Kernel Inception Distance (KID) (Bińkowski et al., 2018) are among the most widely used evaluation metrics for generative image models. Both compute distributional distances in a feature space extracted by InceptionV3 (Szegedy et al., 2016), a classifier pre-trained on ImageNet. This choice of feature extractor creates three structural failure modes when applied to medical image evaluation.

Domain mismatch InceptionV3 was trained to classify natural images into everyday categories such as animals, objects, and scenes, and its internal representations are therefore tuned to natural-image statistics. Consequently, it is insensitive to clinically relevant features in medical images, for example tissue texture heterogeneity in MRI, Hounsfield unit distributions in CT, vascular branching patterns in retinal fundus photography, and ground-glass opacity patterns in chest radiography (Kelkar et al., 2023; Skandarani et al., 2023). Kelkar et al. (2023) showed that state-of-the-art GANs can obtain favorable FID scores on medical images while failing to preserve per-image statistics that are meaningful for downstream clinical tasks. Recent work by Woodland et al. (2024), evaluating sixteen generative models across four medical imaging modalities, found that ImageNet-based feature extractors can be more consistent with human expert judgment than single-modality medical encoders. Backbone domain specificity therefore does not by itself determine metric reliability, which is why the encoder evaluated here is pre-trained across multiple radiology modalities rather than one.

Aggregation over heterogeneous populations FID and KID reduce the distribution of generated images to a single scalar, which can hide heterogeneous structure in the generated sample space. In our brain MRI experiments, a DDPM sits close to the real distribution on a fidelity measure ($\text{MMD}^2 = 0.073$ against a reference whose held-out real counterpart reads 0.000) while covering only 38% of that distribution’s support (Section 4.1). The images it produces are individually plausible; it simply does not produce enough different ones. A favorable distributional score therefore does not imply that a generator covers the real distribution, and no single scalar can separate the two failure modes, because they are not points on one dimension.

Absence of per-image and per-scale interpretability Neither FID, KID, nor manifold-based metrics (Kynkäänniemi et al., 2019; Naeem et al., 2020a) provide per-image scores or a decomposition of distributional shift across clinical domains. A practitioner cannot determine from a single-scalar FID value whether a generated dataset suffers from global intensity drift, geometric distortion, or semantic incoherence, nor identify which individual images are safe to use downstream. This interpretability gap obstructs the deployment of generative models in medical imaging pipelines.

Pixel-level metrics such as Structural Similarity Index (SSIM) (Wang et al., 2004) and Peak signal-to-noise ratio (PSNR) are widely used in medical image synthesis evaluation (Dohmen et al., 2025), yet both carry well-documented limitations. SSIM is insensitive to blurring, unreliable under intensity shifts in unnormalized images, and shows poor correlation with human radiological perception for images containing subtle pathological findings (Breger et al., 2025; Dohmen et al., 2025). PSNR is highly sensitive to normalization choices and inconsistent across different intensity ranges, and prior work recommends against its use in medical image quality assessment (Dohmen et al., 2025). More broadly, Deo et al. (2025) demonstrated that nearly all standard upstream metrics are insensitive to localized, clinically relevant morphological alterations such as distorted tumor boundaries, and that metric rankings of generative models do not align with downstream segmentation performance. Manifold-based metrics (Kynkäänniemi et al., 2019; Naeem et al., 2020a) partially address distributional diversity collapse but carry substantial computational overhead that limits their use in training-time monitoring workflows.

Table 1: Unified benchmark of evaluation metrics for generative medical image models. The M3-Score reports three axes read from blocks fixed *a priori* by representational position (fidelity L12, memorization L9, coverage L4); the fidelity axis additionally carries a native permutation p -value and bootstrap CI (Section 1.2). Each axis is validated against the perturbation it must detect; the coverage axis uses the real-radius estimator of Naeem et al. (2020a), for the reason established in Section 4.5.

Category	Metric	Dir.	Range	Description
Distributional	FID (Heusel et al., 2017)	↓	$[0, \infty)$	Fréchet distance between InceptionV3 feature distributions of real and generated images.
	KID (Bińkowski et al., 2018)	↓	$[0, \infty)$	Unbiased kernel-based distributional distance; more stable than FID at small sample sizes.
Perceptual	SSIM (Wang et al., 2004)	↑	$[0, 1]$	Structural similarity combining luminance, contrast, and structure terms.
	PSNR	↑	$[0, \infty)$	Peak signal-to-noise ratio; sensitive to normalization choices.
	MS-SSIM	↑	$[0, 1]$	Multi-scale extension of SSIM for perceptual consistency at varying resolutions.
	LPIPS (Zhang et al., 2018)	↓	$[0, \infty)$	Learned perceptual image patch similarity using deep visual embeddings.
	IS (Salimans et al., 2016)	↑	$(0, \infty)$	Inception Score; measures sample diversity and classifier confidence.
Manifold	α -Precision (Kynkäänniemi et al., 2019; Alaa et al., 2022)	↑	$[0, 1]$	Fraction of generated samples lying within the real data manifold (fidelity).
	β -Recall (Kynkäänniemi et al., 2019; Alaa et al., 2022)	↑	$[0, 1]$	Fraction of real manifold covered by generated samples (diversity).
	Authenticity	↑	$[0, 1]$	Evaluates whether generated samples avoid memorization of training images.
	Coverage (Naeem et al., 2020b)	↑	$[0, 1]$	Fraction of real samples represented by at least one generated neighbor.
	Density (Naeem et al., 2020b)	↑	$[0, \infty)$	Local concentration of generated samples near the real data manifold.
M3-Score (ours)	M3 Fidelity Axis	↓	$[0, \infty)$	Unbiased multi-bandwidth RBF MMD ² between RadioDINO-s16 final-layer (L12) features of real and generated images, with a native permutation p -value and bootstrap CI.

continued on next page

Category	Metric	Dir.	Range	Description
	M3 Memorization Axis	↓	[0, 1]	Fraction of generated images whose nearest real neighbor at L9 is closer than the median within-real neighbor gap; flags near-duplicate leakage.
	M3 Coverage Axis	↑	[0, 1]	Fraction of real images having at least one generated image inside their own k -NN radius at L4, with radii estimated on the real set (Naeem et al., 2020a); falls monotonically under mode drop at every encoder block (Section 4.5).

1.2. The M3-Score: Design Rationale

We propose the **Medical Multi-axis Maximum Mean Discrepancy (M3-Score)**, an interpretable evaluation framework for generative radiology image models. The M3-Score rests on three design decisions, each motivated by the limitations identified above.

Domain-specific feature extraction via RadioDINO Rather than relying on InceptionV3 features calibrated to natural-image statistics, the M3-Score extracts frozen Vision Transformer (ViT) embeddings from RadioDINO (Pérez-García et al.), a self-supervised encoder pre-trained on radiology images via DINOv2-style distillation (Oquab et al., 2023). Because RadioDINO encodes textural and structural invariances that are clinically relevant in radiology, including tissue boundary sharpness, organ morphology regularity, and pathological texture heterogeneity, the M3-Score is applicable across radiology modalities (CT, MRI) without backbone retraining. We use the compact **RadioDINO-s16** variant (ViT-S/16, 12 blocks, 384-d embeddings). Holding the scoring rule fixed and varying only the feature space (Section 4.3), it separates real brain MRI from in-domain DDPM output at ROC-AUC = 0.819, where the InceptionV3 space underlying FID (0.555) and the CLIP space underlying CMMD (0.582) are close to chance. Claims are restricted to the radiology domain: features for retinal fundus or other non-radiology modalities fall outside the backbone’s training distribution and M3 values there are not directly comparable (Section 4.3).

Three a-priori axes instead of one tuned scalar Distinct generation failures manifest at distinct representational depths, so M3 reports three separately defined quantities rather than a single weighted scalar. Each axis reads from a block fixed *before any test-set result is observed*, by representational position alone: the **fidelity** axis is an unbiased multi-bandwidth RBF MMD² (Gretton et al., 2012) at the final block (L12, the deepest semantic representation, analogous to FID’s use of the InceptionV3 penultimate activation); the **memorization** axis uses nearest-neighbor distances at 75% depth (L9), where features retain instance-level variation that the final layer collapses; and the **coverage** axis measures, at 33% depth (L4) where spatial and structural diversity is encoded, the fraction of real images with a generated neighbour inside their own k -NN radius (Naeem et al., 2020a). The coverage axis is estimated with real-set rather than generated-set radii, for a structural reason developed in Section 3.3 and validated in Section 4.5: the more familiar k -NN recall estimator (Kynkäänniemi et al., 2019) fits its neighbourhoods on the set that a mode-drop probe subsamples, and consequently moves in the wrong direction at every encoder depth. No layer, threshold, or kernel bandwidth in M3 is selected on test-set outcomes; a post-hoc CKA ablation (Section 4.9) independently confirms that the final layer maximizes real-versus-generated discriminability for the fidelity axis, but the axis assignment does not depend on that ablation.

Native statistical reporting For the fidelity axis, M3 reports a permutation-test p -value and a bootstrap 95% confidence interval. FID, KID, and CMMD provide neither. On $N = 500$ images the fidelity evaluation runs in approximately 8 s on a single consumer GPU, making it viable as a checkpoint monitor during training.

1.3. Contributions

The contributions of this paper are as follows.

1. **Three-axis M3 formulation with a-priori depths.** A radiology-domain evaluation framework reporting fidelity (final-block unbiased RBF MMD²), memorization (mid-deep nearest-neighbor distances) and coverage (real-radius manifold coverage at one third depth) as three separately defined quantities. Each depth is fixed by representational position before any test result is seen. We claim only that this eliminates selection-on-outcome bias; we do *not* claim these depths are optimal, and Section 5.1 states what would be required to establish that.
2. **A validated coverage axis, with the failed estimator reported as a negative control.** Coverage specified as k -NN manifold recall (Kynkäänniemi et al., 2019) fails its own mode-drop check at *every* encoder block, in the wrong direction. We show the cause is structural rather than representational: recall fits its hyperspheres on the generated set, so subsampling that set widens them and admits more real points. Substituting the real-radius Coverage estimator (Naeem et al., 2020a), whose radii mode drop cannot move, yields $\rho = -1.00$ at all twelve blocks. We report both, because a sanity check that nothing fails is not evidence of anything (Section 4.5).
3. **Subject-level reference construction.** Neighbouring slices of one volume are near-duplicates, so a reference of N slices taken in filename order can span a handful of patients and is not a sample of N independent images. Every reference in this paper is drawn round-robin across subjects. Under this protocol a held-out real probe correctly registers no shift; the practice is standard for discriminative splits (Varoquaux and Cheplygina, 2022; Glocker et al., 2019) and we adopt it unchanged for generative evaluation.
4. **Statistical reporting for the fidelity axis, honestly scoped.** Each fidelity estimate carries a permutation-test p -value and a bootstrap 95% confidence interval, neither of which FID, KID or CMMD provide. We report these descriptively. The permutation null is exact equality of the two distributions, which every usable generator rejects, so a rejection is not evidence of quality; we make no model-selection claim from it (Section 4.10).
5. **An observed CMMD rank inversion on medical OOD ordering.** We show that CMMD (CLIP ViT-L/14 features) *rank-inverts* a weak same-modality MRI generator and a full cross-modality CT shift, ranking the CT as closer to real MRI (paired $\Delta = -0.039$, 95% CI $[-0.059, -0.021]$, excluding zero), whereas the M3 fidelity axis, and InceptionV3/FID, order them correctly. To our knowledge this specific inversion has not been documented before. We report it as an empirical observation in this setting, not as a general property of CMMD: with a single dataset and one generator pair, and with CLIP’s ViT-L/14 architecture confounded against RadioDINO-s16’s ViT-S/16, we cannot separate the contribution of CLIP’s text-supervised objective from that of its scale or training distribution. Disentangling these requires a CLIP model matched in architecture.
6. **Per-axis validation and lesion specificity.** Each axis is validated against a perturbation it must respond to: near-duplicates injected at known rates for memorization, progressive mode drop for coverage, and a severity ladder for fidelity. Using *real* expert glioma masks with area-matched mirrored and texture-matched controls, M3 is 1.4–1.8 $\times$ more responsive to tumor-region than to control-region perturbation where an Inception-MMD baseline is not ($\sim 1.05\times$); a kernel-controlled baseline attributes the effect to the backbone rather than the kernel. The effect is a *specificity* to subtle diagnostic-region degradation, not a raw-sensitivity advantage, and it is individually significant only in the upper part of the noise sweep. We also report that FID is the more repeatable statistic at every fixed N below 500, and that its expected value nonetheless drifts by 2.52 $\times$ across that range while M3’s drifts by 1.05 $\times$.
7. **Open, protocol-harmonized pipeline.** A single canonical configuration (RadioDINO-s16, $N = 500$ drawn across subjects, multi-bandwidth RBF, U-statistic MMD²) is applied across every experiment, with acceptance criteria fixed before each validation run, and released code and layer caches.

1.4. Scope

The primary validation domain is axial brain MRI slices from the BraTS dataset (Menze et al., 2014; Bakas et al., 2018), chosen for its established benchmark status, large corpus size (38,781 slices), and multi-pathology diversity. Cross-domain experiments additionally use chest/lung CT (LIDC-style) and a non-radiology modality (retinal fundus) to probe the limits of the backbone.

We explicitly scope M3 to the **radiology (CT/MRI) domain** in which RadioDINO was trained. This is a deliberate restriction, not an under-statement: our cross-modality experiment (Section 4.3) shows that M3 ranks retinal fundus images as *further* from brain MRI than chest CT is, which is correct under a radiology-domain ontology (CT $\approx$ MRI $\gg$ fundus) but means raw M3 values are not comparable across the radiology boundary. Applying M3 to a new *radiology*

modality requires only a paired real-vs-generated set and the frozen encoder; applying it outside radiology would require a different backbone.

The M3-Score complements rather than replaces FID (Heusel et al., 2017) and SSIM (Wang et al., 2004): reporting M3 alongside these standard metrics adds a domain-calibrated fidelity axis with native significance testing plus memorization and coverage axes. Method limitations, including the cases where M3 does not outperform existing metrics, are discussed in Section 5.

1.5. Paper Organization

Section 2 reviews related work on generative medical image synthesis and evaluation metrics. Section 3 describes the methodology, including the canonical three-axis a-priori formulation and the legacy post-hoc validation machinery. Section 4 reports experimental results. Section 5 discusses findings, limitations, and the radiology-domain scope. Section 6 concludes.

2. Related Work

Evaluating generative image models is a long-standing open problem. The field has evolved through three broad phases: hand-crafted pixel-level metrics, learned distributional metrics anchored to natural-image classifiers, and, most recently, task-aware or domain-specific metrics motivated by the inadequacy of general-purpose feature extractors for medical imaging. Each phase is reviewed below.

2.1. Phase 1: Pixel-Level and Structural Metrics

The earliest quantitative metrics for image quality operate at the pixel level. PSNR measures reconstruction fidelity as the logarithm of the ratio between the maximum possible signal and the mean squared error, making it sensitive to normalization choices and inconsistent across intensity ranges. Wang et al. (2004) introduced the SSIM, which partially addresses these limitations by jointly evaluating luminance, contrast, and structural consistency within local windows. SSIM became the de facto paired-image metric in medical image synthesis (Dohmen et al., 2025) and remains widely reported alongside distributional metrics. However, Breger et al. (2025) and Dohmen et al. (2025) have both shown that SSIM correlates poorly with radiological perception for images containing subtle pathological findings, and that PSNR is unreliable under the normalization conventions common in medical imaging. Zhang et al. (2018) introduced the Learned Perceptual Image Patch Similarity (LPIPS) metric, which replaces hand-crafted structural terms with a calibrated distance in a pretrained deep feature space. While LPIPS aligns better with human judgment than PSNR or SSIM on natural images, it inherits the ImageNet bias of its backbone and is designed for paired comparisons, limiting its applicability to unpaired generative model evaluation.

2.2. Phase 2: Distributional Metrics and Modern Embeddings

2.2.1. Distributional Metrics Based on Inception Features

The dominant paradigm for evaluating unpaired generative models originated with the Inception Score (IS) introduced by Salimans et al. (2016), which measures the sharpness and diversity of generated images by applying a pretrained InceptionV3 classifier (Szegedy et al., 2016) to the generated set and computing the KL divergence between the conditional and marginal label distributions. IS is computed on generated images alone, with no reference to real data, and therefore cannot detect mode collapse or distribution shift.

FID, introduced by Heusel et al. (2017) as part of a two-time-scale GAN training framework, addressed this limitation by computing the Fréchet distance between multivariate Gaussians fitted to the InceptionV3 feature distributions of real and generated images. FID rapidly became the primary benchmark metric across the generative modeling literature and remains the most reported scalar in medical image synthesis evaluations. However, FID carries three structural limitations that are particularly acute in the medical setting: it assumes Gaussian-distributed activations (which is violated by ReLU features), it is a biased estimator whose bias scales inversely with sample size, and its backbone encodes natural-image rather than clinical image statistics (Kelkar et al., 2023; Skandarani et al., 2023).

KID, proposed by Bińkowski et al. (2018), addressed the bias and distributional assumptions of FID by replacing the Fréchet distance with an unbiased Maximum Mean Discrepancy (MMD) estimator using a polynomial kernel over the same InceptionV3 features. KID admits an analytic unbiased estimator, is more reliable than FID at the small sample sizes typical in medical imaging, and does not presuppose a parametric distributional form. The underlying MMD framework was formalized by Gretton et al. (2012) as a nonparametric two-sample test whose statistic is the largest difference in expectations over functions in a reproducing kernel Hilbert space. KID retains the InceptionV3

backbone and therefore shares the domain mismatch problem of FID, though its statistical properties are considerably stronger.

2.2.2. Moving Beyond Inception: Richer Embeddings and Kernels

Jayasumana et al. (2024) proposed CLIP Maximum Mean Discrepancy (CMMD) as a direct replacement for FID. CMMD computes an unbiased Gaussian RBF kernel MMD over CLIP (Radford et al., 2021) embeddings, which encode richer semantic content than InceptionV3 because CLIP is trained on diverse image-text pairs at scale. The authors demonstrated empirically that FID contradicts human raters, fails to reflect gradual quality improvement in iterative text-to-image models, and can paradoxically improve when progressive distortions are applied; CMMD behaves monotonically and correctly under all three conditions. CMMD is distribution-free, sample-efficient, and provides an unbiased estimate. However, CLIP embeddings are still derived from natural image-text pairs and do not encode the anatomical and pathological structure of medical images.

2.3. Phase 3: Manifold-Based, Domain-Specific, and Radiology-Aware Metrics

2.3.1. Manifold-Based Fidelity and Diversity Metrics

A complementary line of work decouples the fidelity and diversity aspects of generative model quality that FID collapses into a single scalar. Kynkäänniemi et al. (2019) introduced improved precision and recall metrics that form explicit, non-parametric approximations to the real and generated data manifolds using k -nearest neighbor sets: Precision measures the fraction of generated samples that fall within the real manifold, and Recall measures the fraction of the real manifold covered by generated samples. Naeem et al. (2020b) subsequently showed that the Kynkäänniemi precision and recall metrics are unreliable in the presence of outliers and fail to detect identical distributions; they proposed Density and Coverage as more robust alternatives that use overlapping manifold hyperspheres rather than strict membership. These manifold metrics provide complementary diagnostic information but require dense sampling to form reliable manifold estimates, making them computationally prohibitive at training-time monitoring scales.

2.3.2. Domain-Specific Metrics for Medical Image Evaluation

The inadequacy of ImageNet-pretrained feature extractors for medical image evaluation has motivated a small but growing body of domain-specific metrics. Kelkar et al. (2023) demonstrated that state-of-the-art GANs can attain favorable FID scores on medical images while failing to preserve per-image statistics that matter for downstream clinical tasks. Deo et al. (2025) showed that nearly all upstream metrics are insensitive to localized morphological distortions such as tumor boundary deformation, and that metric rankings do not align with downstream segmentation performance.

Konz et al. (2026) introduced the Fréchet Radiomic Distance (FRD), the most direct precursor to the M3-Score in the medical imaging literature. FRD replaces deep learned features with standardized radiomic features from PyRadiomics, computing the Fréchet distance between Gaussian-fitted radiomic distributions of real and generated images. FRD was shown to outperform FID, KID, and CMMD across OOD detection, image-to-image translation evaluation, and unconditional generation quality, with additional advantages of stability at small sample sizes and correlation with radiologist-perceived quality; it has been published in Medical Image Analysis (Konz et al., 2026).

The M3-Score differs from FRD in three respects. First, it uses deep learned features from a radiology-pretrained ViT (RadioDINO (Pérez-García et al.)) rather than hand-crafted radiomic features, capturing higher-order structural and semantic representations not available to first-order or texture-based descriptors. Second, rather than a single scalar, M3 reports two separately-defined axes read from *fixed a-priori* layers (fidelity at L12 and memorization at L9), which decouple distinct failure modes. Third, the fidelity axis is an *unbiased* multi-bandwidth RBF MMD² reported with a native permutation p -value and bootstrap CI, whereas FRD computes a Fréchet (Gaussian-moment) distance without a built-in significance test. We do not report an FRD comparison here: its reference implementation depends on `pyradiomics`, which does not build on the Python version used for this work, and we prefer to omit a baseline rather than quote one we could not recompute under the reference protocol of Section 3.2. A radiomic-feature baseline remains a useful comparison and we note it as future work. One implementation caveat applies: PyRadiomics declines to apply the Laplacian-of-Gaussian filter to the *wavelet-decomposed* sub-images, which are half-resolution (120×120 and 128×128 for our 240×240 real and 256×256 generated slices) and fall below its size threshold. The LoG-on-wavelet feature subset is therefore skipped; FRD is still computed from the remaining first-order, texture, and LoG-on-original classes. This narrows the feature set slightly relative to the full FRDv1 specification but does not prevent the comparison.

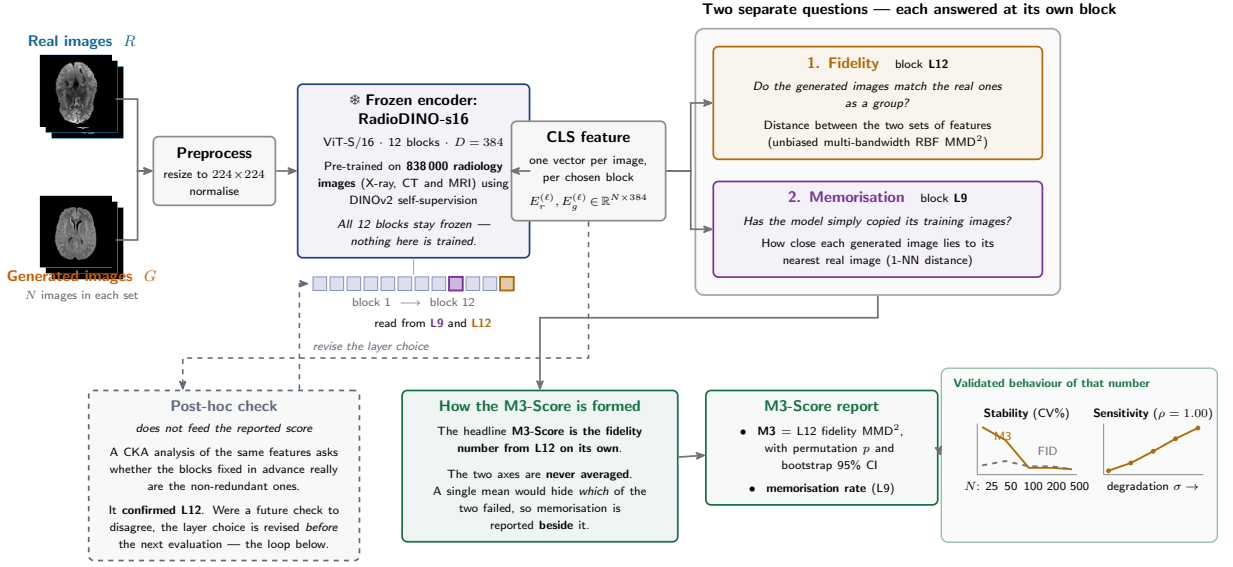


Figure 1: Overview of the M3-Score evaluation pipeline. The real set R and generated set G (N images each) pass through a single frozen RadioDINO-s16 encoder (ViT-S/16, pre-trained on 838,000 radiology images with DINOv2 self-supervision); nothing is trained or fine-tuned. One CLS-token feature vector per image is read from each of two *a-priori* layers: L12 asks whether the generated set matches the real set as a group (fidelity) and L9 asks whether the model has copied its training images (memorization). The two axes are never averaged: the headline M3-Score is the L12 fidelity MMD² alone, reported with a permutation p -value and a bootstrap 95% CI, with the memorization rate beside it. The dashed path is an optional post-hoc CKA check that never contributes to the reported score. Inset plots summarize the validated behavior of the reported number (Sections 4.8 and 4.2).

2.3.3. Self-Supervised Radiology Encoders

The M3-Score’s feature extractor is RadioDINO, introduced by Pérez-García et al., a family of Vision Transformer encoders pre-trained on over 838,000 radiology images spanning multiple modalities and anatomical regions using the DINOv2 (Oquab et al., 2023) self-supervised distillation objective; we adopt the compact ViT-S/16 variant (RadioDINO-s16). RadioDINO was trained without text supervision, using image-level and patch-level objectives, and was shown to match or exceed the performance of text-supervised biomedical encoders on classification, segmentation, and report generation benchmarks. The absence of text supervision and the breadth of the radiology training corpus make RadioDINO’s intermediate ViT representations more likely to encode the anatomical and pathological structure relevant to generation quality assessment than either CLIP or InceptionV3 features (Pérez-García et al.).

Woodland et al. (2024) conducted a large-scale study evaluating sixteen generative models across four medical imaging modalities and found that the choice of backbone substantially affects metric consistency with human expert judgment, motivating the use of a purpose-built multi-modality radiology encoder rather than a single-modality or general-purpose backbone.

Three recurring gaps emerge from the work reviewed above: (a) dependence on general-purpose encoders misaligned with radiology semantics, (b) collapse of evaluation into a single scalar that conflates distinct failure modes (fidelity, memorization, coverage), and (c) absence of native significance testing and per-image interpretability. M3-Score addresses all three by building on a purpose-built radiology encoder (RadioDINO-s16), reporting separate fidelity, memorization and coverage axes read from blocks fixed *a priori* by representational depth, and accompanying the fidelity axis with a permutation p -value and a bootstrap confidence interval. Section 3 develops the construction.

3. Methodology

The superseded adaptive design (CKA pruning and entropy-stability-uniqueness weighting) is deferred to Appendix A.

Algorithm 1 M3-Score (canonical three-axis, a-priori depths)

Require: Real set $\mathcal{X}_r$ (drawn across subjects, §3.2), generated set $\mathcal{X}_g$, frozen encoder Φ , a-priori depths $\ell_{\text{fid}}=12, \ell_{\text{mem}}=9, \ell_{\text{cov}}=4$

Ensure: Fidelity ($\widehat{\text{MMD}}^2, p, [\text{CI}]$), memorization rate, coverage

- 1: **for** $\ell \in \{12, 9, 4\}$ **do**
- 2: $E_r^{(\ell)} \leftarrow \text{CLS}(\Phi, \mathcal{X}_r, \ell)$;
- 3: $E_g^{(\ell)} \leftarrow \text{CLS}(\Phi, \mathcal{X}_g, \ell)$ ▷ CLS tokens; §3.4
- 4: **end for**
- 5: // Fidelity axis (L12)
- 6: $\widehat{\text{MMD}}^2 \leftarrow \text{MMD2U}(E_r^{(12)}, E_g^{(12)})$ ▷ unbiased multi-bandwidth RBF; Alg. 3
- 7: $p \leftarrow \text{PERMTEST}(E_r^{(12)}, E_g^{(12)}; B=500)$
- 8: $[\text{CI}] \leftarrow \text{BOOTSTRAP}(E_r^{(12)}, E_g^{(12)}; B=500)$
- 9: // Memorization axis (L9)
- 10: $\theta \leftarrow \text{median within-real 1-NN distance}$
- 11: $\text{mem} \leftarrow \frac{1}{N} \sum_j \mathbb{1} \left[\min_i \|E_{g,j}^{(9)} - E_{r,i}^{(9)}\|_2 < \theta \right]$
- 12: // Coverage axis (L4), radii from the REAL set
- 13: $r_i \leftarrow \text{NND}_k(E_{r,i}^{(4)}, E_r^{(4)})$ for each i ▷ $k=5$; within-real, excluding self
- 14: $\text{cov} \leftarrow \frac{1}{N} \sum_i \mathbb{1} \left[\min_j \|E_{r,i}^{(4)} - E_{g,j}^{(4)}\|_2 \leq r_i \right]$
- 15: **return** $\{\widehat{\text{MMD}}^2, p, [\text{CI}]\}$, mem, cov

3.1. Notation and Pipeline Overview

Let $\mathcal{X}_r = \{x_i\}_{i=1}^N$ denote a reference set of N real radiology images and $\mathcal{X}_g = \{\hat{x}_j\}_{j=1}^N$ a set of N images produced by a generative model under evaluation. Let Φ denote the frozen RadioDINO-s16 encoder, which maps an input image to a sequence of intermediate representations across the $L = 12$ transformer blocks. The canonical M3-Score reports three axes computed in the following stages, detailed in Section 3.3 and summarized in Algorithm 1. (A superseded adaptive path, retained only for the post-hoc validation of Section 4.9, is *not* part of the canonical metric.)

1. **A-priori layer read-out.** Features are extracted from three fixed layers, $\ell_{\text{fid}} = 12, \ell_{\text{mem}} = 9$, and $\ell_{\text{cov}} = 4$, chosen by representational depth before any test outcome is observed. No layer is selected adaptively.
2. **Feature extraction.** For each axis layer ℓ , CLS-token features yield embedding matrices $E_r^{(\ell)}, E_g^{(\ell)} \in \mathbb{R}^{N \times D}$ for the real and generated sets ($D = 384$).
3. **Fidelity axis.** An unbiased multi-bandwidth RBF MMD^2 U-statistic quantifies the distributional divergence between $E_r^{(12)}$ and $E_g^{(12)}$, reported with a permutation p -value and a bootstrap 95% CI.
4. **Memorization axis.** Nearest-neighbor distances at L9 give the memorization rate. The legacy CKA-pruning step (Algorithm 5) is run only when the optional post-hoc validation of Section 4.9 is desired.

3.2. Reference Set Construction

Every distributional metric compares a generated set against a finite set of real images, and its value is a property of that reference as much as of the generator. In volumetric medical imaging the reference requires care that natural-image benchmarks do not. Adjacent axial slices of one volume are near-duplicates of one another, so a reference of N slices is not a sample of N independent images: its statistical content is governed by the number of distinct *subjects* it contains. A reference assembled the most convenient way, by taking the first N files in sorted order, is worse still, because public collections order filenames by subject identifier; such a draw selects a contiguous block of a few patients, which on BraTS also means a small number of contributing institutions and therefore a particular scanner and protocol signature.

We therefore draw every real set in this paper round-robin across subjects: subjects are shuffled, then one slice is taken from each in turn before any second slice is taken from any, so the draw spans as many distinct patients as its

size allows. Where a real-vs-real comparison is reported, the two blocks are drawn from *disjoint* subject pools. The canonical reference is $N = 500$ slices, and under this procedure it spans 500 distinct subjects.

This is the convention discriminative medical imaging already uses. Subject-level rather than image-level splits are standard there, because slices from one subject otherwise leak between train and test (Varoquaux and Cheplygina, 2022), and multi-site collections are known to carry scanner signatures recoverable by a classifier (Glocker et al., 2019). The reference set in a generative study occupies the structural position that a test split occupies in a discriminative one, so we apply the same rule. We report the subject count alongside N for every set we evaluate, and recommend the practice generally: the two numbers are not interchangeable, and only one of them is usually given.

3.3. Canonical Three-Axis Formulation (A-Priori Depths)

The canonical M3-Score reports three axes, each read from a fixed block of the RadioDINO-s16 encoder ($L = 12$ blocks). The block assignments are chosen by representational position *before any evaluation outcome is observed*, and are frozen configuration constants:

$$\ell_{\text{fid}} = 12, \quad \ell_{\text{mem}} = 9, \quad \ell_{\text{cov}} = 4. \quad (1)$$

The justification is structural, not empirical: $\ell_{\text{fid}} = 12$ is the deepest semantic block (the standard choice for distributional distance, mirroring FID’s use of the InceptionV3 penultimate activation); $\ell_{\text{mem}} = 9$ sits at 75% depth, where features retain instance-level variation useful for near-duplicate detection that the final block collapses; $\ell_{\text{cov}} = 4$ sits at 33% depth, where spatial and structural diversity is encoded. No layer, threshold, or kernel bandwidth is selected on test-set outcomes. These are *structural priors*, and we are explicit about what that does and does not buy. It buys freedom from selection-on-outcome bias: no depth in this paper was chosen because it produced a better result. It does not buy optimality, and we make no optimality claim for any of the three. The fidelity depth L12 is the one block a CKA pruning procedure never discards, at any redundancy threshold (Section 4.9); that is a statement about non-redundancy, not about optimality. L9 and L4 rest on representational position alone. Section 5.1 states what a depth-optimality argument would require.

Fidelity axis Let $E_r^{(12)}, E_g^{(12)} \in \mathbb{R}^{N \times D}$ be the CLS-token features at L12. The fidelity score is the unbiased multi-bandwidth RBF MMD² U-statistic between them (Section 3.5), with bandwidths set by the median heuristic on the pooled sample. We report it with: a permutation-test p -value (pool $E_r \cup E_g$, $B = 500$ label shuffles, one-sided $H_0 : \text{MMD}^2 = 0$); a null-standardized separation $Z = (\widehat{\text{MMD}}^2 - \mu_0)/\sigma_0$ against the permutation null (μ_0, σ_0); and a bootstrap 95% CI (500 resamples of r and g independently with replacement). We caution that Z is reported only as a separation magnitude, *not* as a Gaussian z -score: under H_0 the unbiased MMD² null variance σ_0^2 shrinks as $O(1/N)$ while the observed statistic converges to a positive constant, so Z grows approximately linearly with N . The permutation null is moreover strongly non-Gaussian, so Z does not map to a normal tail probability; inference rests on the permutation p -value and the bootstrap CI.

Memorization axis Using L9 features, for each generated image we compute its 1-NN L2 distance to the real set, $d_j = \min_i \|E_{g,j}^{(9)} - E_{r,i}^{(9)}\|_2$. Let θ be the median within-real 1-NN distance (excluding self). The memorization rate is $\frac{1}{N} \sum_j \mathbb{1}[d_j < \theta]$: the fraction of generated images closer to a real image than a typical real-real neighbor gap, flagging candidate near-duplicates.

Coverage axis Using L4 features, coverage is the fraction of *real* images that have at least one generated image inside their own k -NN radius ($k = 5$), where that radius is estimated within the real set (Naeem et al., 2020a):

$$\text{Cov} = \frac{1}{N} \sum_{i=1}^N \mathbb{1} \left[\min_j \|E_{r,i}^{(4)} - E_{g,j}^{(4)}\|_2 \leq \text{NND}_k(E_{r,i}^{(4)}) \right], \quad (2)$$

with $\text{NND}_k(\cdot)$ the distance to the k -th nearest neighbour within the real set. Low coverage means the generator leaves regions of the real manifold unpopulated, i.e. mode dropping.

The choice of estimator is deliberate and is made on structural grounds. The more familiar k -NN manifold recall (Kynkäänniemi et al., 2019) answers a superficially identical question but places its hyperspheres on the *generated* points, with radii equal to their k -NN distances within the generated set. Those radii are not fixed quantities:

removing generated samples spreads the survivors apart, enlarges their k -NN distances, and widens the spheres, so recall can *increase* precisely when generated diversity is destroyed. Equation (2) places the spheres on the real points instead. Their radii depend only on the real reference, which mode drop does not touch, so removing generated samples can only leave real points uncovered: coverage is monotone non-increasing under mode drop by construction. Section 4.5 measures both estimators and confirms that the two behave as this argument predicts, at every encoder block.

The adaptive CKA-pruning and entropy-stability-uniqueness machinery, deferred to Appendix A, is retained only as an *optional, post-hoc validation* tool (it independently confirms L12 is the most discriminative fidelity layer, Section 4.9); it is not part of the canonical metric and never selects the reported axes.

An earlier *adaptive* design (CKA-based layer pruning followed by an entropy-stability-uniqueness weighting of the layer distances) was abandoned in favor of the fixed a-priori layers of Section 3.3, so that no layer, threshold, or weight is ever selected on test outcomes. The CKA layer-selection step it introduced is retained purely as a post-hoc check and is specified in Appendix A (Algorithm 5); that check confirms L12 as the single most discriminative fidelity layer. The abandoned aggregation and weighting rules are not reproduced here, as no reported result uses them.

3.4. Feature Extraction via Frozen RadioDINO

Encoder architecture RadioDINO (Pérez-García et al.) is a family of Vision Transformer encoders (Dosovitskiy et al., 2021) pre-trained on 838,000 radiology images spanning chest radiography, computed tomography, and magnetic resonance imaging from three countries, using the DINOv2 (Oquab et al., 2023) self-supervised distillation objective. We use the compact **RadioDINO-s16** variant (ViT-S/16), selected for its leading OOD discriminability among six candidate backbones (Section 4.3). The encoder has $L = 12$ transformer blocks, each with hidden dimension $D = 384$ and $H = 6$ attention heads. An input image of resolution 224×224 is partitioned into 196 non-overlapping patches of size 16×16 ; a learnable CLS token is prepended, giving a sequence of length $N_{\text{seq}} = 197$. The encoder weights are fixed throughout all M3-Score computations.

Per-image feature: CLS token (canonical) At transformer block ℓ , the encoder produces a hidden-state tensor $\mathbf{H}^{(\ell)} \in \mathbb{R}^{B \times N_{\text{seq}} \times D}$ (and, for backbones that expose it, an attention tensor $\mathbf{A}^{(\ell)} \in \mathbb{R}^{B \times H \times N_{\text{seq}} \times N_{\text{seq}}}$), where B is the mini-batch size and $b \in \{1, \dots, B\}$ indexes an image within the batch. For the canonical RadioDINO-s16 backbone (loaded via `timm(Wightman, 2019)`), the per-image feature at block ℓ is the CLS token $f_b^{(\ell)} = \mathbf{H}_{b,0,:}^{(\ell)} \in \mathbb{R}^D$, consistent with Section 3.3 and with the standard practice of reading a ViT’s aggregated representation from its class token. This is the configuration used for all reported M3-Score results, and it is what Fig. 1 depicts. Stacking $f_b^{(\ell)}$ over the N images of $\mathcal{X}$ yields the embedding matrix $E^{(\ell)} \in \mathbb{R}^{N \times D}$.

Attention-aware pooling (alternative for transformers backbones) For HuggingFace transformers (Wolf et al., 2020) ViT backbones that additionally expose per-head attention (e.g. Microsoft rad-dino, DINOv2), M3 supports an *attention-aware* pooling as an alternative to the CLS token. It is evaluated in the backbone comparison but is *not* used for any RadioDINO-s16 result reported here. Under this alternative, a weighted pooling is applied using the attention distribution that the CLS token assigns to each patch token. Let $\bar{\mathbf{A}}^{(\ell)} \in \mathbb{R}^{B \times N_{\text{seq}} \times N_{\text{seq}}}$ be the head-averaged attention matrix,

$$\bar{\mathbf{A}}_b^{(\ell)} = \frac{1}{H} \sum_{h=1}^H \mathbf{A}_{b,h,:,\cdot}^{(\ell)}. \quad (3)$$

The CLS-to-patch attention weights are defined as

$$\alpha_{b,p}^{(\ell)} = \bar{\mathbf{A}}_{b,0,p+1}^{(\ell)}, \quad p = 1, \dots, N_{\text{seq}} - 1. \quad (4)$$

The index $p + 1$ selects patch positions only; the CLS self-attention weight $\bar{\mathbf{A}}_{b,0,0}^{(\ell)}$ is excluded, since DINOv2-style encoders produce systematically elevated self-attention at the CLS position that would otherwise dominate the pooled

Algorithm 2 Attention-aware feature extraction**Require:** Encoder Φ , image set $\mathcal{X}$, target layer set $\mathcal{L}$, mini-batch size $B = 16$ **Ensure:** $\{E^{(\ell)}\}_{\ell \in \mathcal{L}}$, $E^{(\ell)} \in \mathbb{R}^{|\mathcal{X}| \times D}$

```

1: buf $[\ell] \leftarrow \emptyset$  for all  $\ell \in \mathcal{L}$ 
2: for each mini-batch  $B \subseteq \mathcal{X}$  of size  $\min(B, |\mathcal{X}|)$  do
3:    $B \leftarrow$  apply RadioDINO image processor to  $B$ 
4:    $\{\mathbf{H}^{(\ell)}\}, \{\mathbf{A}^{(\ell)}\} \leftarrow \Phi(B)$   $\triangleright$  all layers, hidden states and attentions
5:   for each  $\ell \in \mathcal{L}$  do
6:      $\bar{\mathbf{A}}^{(\ell)} \leftarrow \frac{1}{H} \sum_{h=1}^H \mathbf{A}_{:,h,:}^{(\ell)}$   $\triangleright (B \times N_{\text{seq}} \times N_{\text{seq}})$ 
7:      $\alpha^{(\ell)} \leftarrow \bar{\mathbf{A}}_{:,0,1}^{(\ell)}$   $\triangleright$  CLS-to-patch weights,  $(B \times (N_{\text{seq}} - 1))$ 
8:      $\tilde{\alpha}^{(\ell)} \leftarrow \alpha^{(\ell)} / (\sum_p \alpha_{:,p}^{(\ell)} + \epsilon)$   $\triangleright \ell_1$ -normalize, Eq. (5)
9:      $\mathbf{P}^{(\ell)} \leftarrow \mathbf{H}_{:,1,:}^{(\ell)}$   $\triangleright$  patch tokens,  $(B \times (N_{\text{seq}} - 1) \times D)$ 
10:     $f^{(\ell)} \leftarrow \sum_p \tilde{\alpha}_{:,p}^{(\ell)} \cdot \mathbf{P}_{:,p,:}^{(\ell)}$   $\triangleright$  weighted sum, Eq. (6),  $(B \times D)$ 
11:    buf $[\ell] \leftarrow \text{buf}[\ell] \cup \{f^{(\ell)}\}$ 
12:  end for
13: end for
14: for each  $\ell \in \mathcal{L}$  do
15:    $E^{(\ell)} \leftarrow \text{concatenate}(\text{buf}[\ell], \text{axis}=0)$ 
16: end for
17: return  $\{E^{(\ell)}\}_{\ell \in \mathcal{L}}$ 

```

vector (Darcet et al., 2024). The weights are ℓ_1 -normalized per image:

$$\tilde{\alpha}_{b,p}^{(\ell)} = \frac{\alpha_{b,p}^{(\ell)}}{\sum_{p'=1}^{N_{\text{seq}}-1} \alpha_{b,p'}^{(\ell)} + \epsilon}, \quad \epsilon = 10^{-8}. \quad (5)$$

The attention-pooled per-image feature vector at layer ℓ for image b is

$$f_b^{(\ell), \text{attn}} = \sum_{p=1}^{N_{\text{seq}}-1} \tilde{\alpha}_{b,p}^{(\ell)} \cdot \mathbf{H}_{b,p+1,:}^{(\ell)} \in \mathbb{R}^D, \quad (6)$$

which replaces the canonical CLS feature $f_b^{(\ell)}$ only when this alternative backbone path is selected. Algorithm 2 states the complete attention-aware procedure, including mini-batch processing and accumulation across batches.

3.5. Per-Layer Unbiased MMD²

Feature normalization RadioDINO intermediate hidden states returned by the transformer blocks have ℓ_2 norms in the range $[10^2, 3 \times 10^2]$ depending on layer depth. Without correction, this arbitrary scale dominates the pairwise Euclidean distances that enter the Gaussian kernel and destabilizes the median-heuristic bandwidth, rendering MMD² estimates incomparable across layers and datasets. To eliminate this dependence on the arbitrary scale of intermediate hidden states, each embedding vector is projected onto the unit sphere prior to kernel evaluation:

$$\hat{e} = \frac{e}{\|e\|_2 + \epsilon}, \quad \epsilon = 10^{-8}. \quad (7)$$

After this projection, $\|\hat{e}\|_2 = 1$ and the inner product $\hat{u}^\top \hat{v}$ equals the cosine similarity of u and v , which lies in $[-1, 1]$. This normalization is applied to the rows of both $E_r^{(\ell)}$ and $E_g^{(\ell)}$ before any kernel computation. The CKA layer selection step (Appendix A) operates on the raw (unnormalized) features, since CKA is scale-invariant and measures representational geometry rather than distributional distance.

Kernel definition The distributional divergence between the unit-normalized real and generated embedding sets is measured by the Maximum Mean Discrepancy (Gretton et al., 2012) under a *multi-bandwidth Gaussian RBF* kernel. For two unit-norm vectors $u, v \in \mathbb{R}^D$,

$$k(u, v) = \frac{1}{|\mathcal{B}|} \sum_{\sigma^2 \in \mathcal{B}} \exp\left(-\frac{\|u - v\|_2^2}{2\sigma^2}\right), \quad (8)$$

the average of five Gaussian kernels whose squared bandwidths

$$\mathcal{B} = \left\{\frac{1}{4}, \frac{1}{2}, 1, 2, 4\right\} \cdot \sigma_{\text{med}}^2 \quad (9)$$

form a geometric ladder around a base scale σ_{med}^2 set by the *median heuristic* (Gretton et al., 2012): σ_{med}^2 is the median of the pairwise squared Euclidean distances over the pooled sample $\hat{R} \cup \hat{G}$ (upper triangle, self-distances excluded). Averaging over a ladder of bandwidths makes the kernel sensitive to distributional differences at several scales simultaneously and removes the need to tune a single bandwidth, while the median anchor adapts the kernel to the geometry of each feature cloud. On the unit sphere, $\|u - v\|_2^2 = 2(1 - u^\top v) \in [0, 4]$, so every kernel value lies in $(0, 1]$ and the resulting MMD² is bounded and directly comparable across datasets, analogous to KID (Bińkowski et al., 2018).

Unbiased estimator Let $\hat{R} \in \mathbb{R}^{N \times D}$ and $\hat{G} \in \mathbb{R}^{N \times D}$ denote the ℓ_2 -normalized real and generated embedding matrices at the fidelity layer. Define the kernel matrices $K_{ij}^{rr} = k(\hat{R}_{i,:}, \hat{R}_{j,:})$, $K_{ij}^{gg} = k(\hat{G}_{i,:}, \hat{G}_{j,:})$, $K_{ij}^{rg} = k(\hat{R}_{i,:}, \hat{G}_{j,:})$ using the multi-bandwidth RBF kernel of Eq. (8) (by linearity, averaging the per-bandwidth U-statistics is equivalent to forming the U-statistic of the averaged kernel). The unbiased MMD² estimator is

$$\widehat{\text{MMD}}_\ell^2 = \frac{1}{N(N-1)} \sum_{i \neq j} K_{ij}^{rr} + \frac{1}{N(N-1)} \sum_{i \neq j} K_{ij}^{gg} - \frac{2}{N^2} \sum_{i,j} K_{ij}^{rg}, \quad (10)$$

clamped below at zero to remove finite-sample negative artifacts; this introduces a small one-sided bias only in the regime where the true MMD² is near zero, and the estimator remains asymptotically unbiased. Unlike FID, Eq. (10) makes no parametric assumption on the distribution of embeddings, is an (asymptotically) unbiased estimator of the true population MMD², and its expectation is zero if and only if the real and generated distributions are identical under k . Algorithm 3 states the computation explicitly; we refer to it as MMD2U where it is invoked from other algorithms.

3.6. Per-Image Score

The population-level score S summarizes distributional divergence across all N generated images. A complementary per-image score enables ranking of individual generated images by their distance from the real distribution, which is absent from FID, KID, and FRD. Consistent with the canonical fidelity axis, the per-image score is defined on the final (L12) layer alone, so it requires no adaptive weighting. For an image $\hat{x}_j \in \mathcal{X}_g$ with ℓ_2 -normalized feature vector $\hat{E}_{g,j}^{(12)}$, the score is the distance to the real centroid

$$s(\hat{x}_j) = \left\| \hat{E}_{g,j}^{(12)} - \hat{\mu}_r^{(12)} \right\|_2, \quad \hat{\mu}_r^{(12)} = \frac{1}{N} \sum_{i=1}^N \hat{E}_{r,i}^{(12)}, \quad (11)$$

where $\hat{\mu}_r^{(12)}$ is the empirical centroid of the ℓ_2 -normalized real L12 embeddings. This is exactly the per-image quantity used in the anomaly-correlation experiment (Section 4.3, Fig. 7). Images with large $s(\hat{x}_j)$ lie further from the real distribution at the semantic (L12) scale. (An unweighted average over the three canonical layers $\{12, 9, 4\}$ is an equally valid variant; we default to L12 for direct correspondence with the fidelity axis.) The centroid $\hat{\mu}_r^{(\ell)}$ is a by-product of the feature extraction step, so no additional computation is required beyond what is performed for S . Algorithm 4 states the ranking procedure.

3.7. Computational Complexity

Time complexity The dominant cost is feature extraction, which requires a forward pass through Φ for all N images. Self-attention at each block costs $O(N_{\text{seq}}^2 \cdot D)$ per image, giving $O(N \cdot N_{\text{seq}}^2 \cdot D)$ total for a single set. The CKA layer

Algorithm 3 Unbiased multi-bandwidth RBF MMD²

Require: Real and generated embedding matrices at a single layer ℓ , $E_r, E_g \in \mathbb{R}^{N \times D}$ (Section 3.1); bandwidth scales $S = \{0.25, 0.5, 1, 2, 4\}$

Ensure: $\widehat{\text{MMD}}^2 \in [0, \infty)$

- 1: $\hat{E}_{r,i,:} \leftarrow E_{r,i,:} / (\|E_{r,i,:}\|_2 + \varepsilon)$; $\hat{E}_{g,j,:} \leftarrow E_{g,j,:} / (\|E_{g,j,:}\|_2 + \varepsilon)$ $\triangleright \ell_2$ -normalize rows, Eq. (7)
- 2: $D_{ij}^{rr} \leftarrow \|\hat{E}_{r,i} - \hat{E}_{r,j}\|_2^2$; $D_{ij}^{gg} \leftarrow \|\hat{E}_{g,i} - \hat{E}_{g,j}\|_2^2$; $D_{ij}^{rg} \leftarrow \|\hat{E}_{r,i} - \hat{E}_{g,j}\|_2^2$
- 3: $\sigma_{\text{med}}^2 \leftarrow \text{median}(\{D_{ij}^U : i < j\})$ $\triangleright$ median heuristic on $\hat{E}_r \cup \hat{E}_g$, Eq. (9)
- 4: $A \leftarrow 0$
- 5: **for** each scale $s \in S$ **do**
- 6: $\gamma \leftarrow 1/(2s\sigma_{\text{med}}^2)$
- 7: $K^{rr} \leftarrow \exp(-\gamma D^{rr})$; $K^{gg} \leftarrow \exp(-\gamma D^{gg})$; $K^{rg} \leftarrow \exp(-\gamma D^{rg})$
- 8: $T_{rr} \leftarrow (\sum_{i,j} K_{ij}^{rr} - \sum_i K_{ii}^{rr}) / (N(N-1))$ $\triangleright$ diagonal excluded
- 9: $T_{gg} \leftarrow (\sum_{i,j} K_{ij}^{gg} - \sum_i K_{ii}^{gg}) / (N(N-1))$
- 10: $T_{rg} \leftarrow (\sum_{i,j} K_{ij}^{rg}) / N^2$
- 11: $A \leftarrow A + (T_{rr} + T_{gg} - 2T_{rg})$
- 12: **end for**
- 13: $\widehat{\text{MMD}}^2 \leftarrow \max(A/|S|, 0)$ $\triangleright$ Eq. (10); clamp removes finite- N negative bias
- 14: **return** $\widehat{\text{MMD}}^2$

Algorithm 4 Per-image quality ranking

Require: Generated L12 embeddings $E_g^{(12)}$, real L12 embeddings $E_r^{(12)}$

Ensure: Index permutation π such that $s_{\pi(1)} \geq s_{\pi(2)} \geq \dots \geq s_{\pi(N)}$

- 1: ℓ_2 -normalize all rows of $E_r^{(12)}$ and $E_g^{(12)}$ $\triangleright$ Eq. (7)
- 2: $\hat{\mu}_r^{(12)} \leftarrow \frac{1}{N} \sum_{i=1}^N \hat{E}_{r,i}^{(12)}$ $\triangleright$ real centroid at L12
- 3: **for** $j = 1, \dots, N$ **do**
- 4: $s_j \leftarrow \|\hat{E}_{g,j}^{(12)} - \hat{\mu}_r^{(12)}\|_2$ $\triangleright$ Eq. (11)
- 5: **end for**
- 6: $\pi \leftarrow \text{argsort}(s_1, \dots, s_N, \text{descending})$
- 7: **return** π

selection step processes N_r images over all L layers and requires $O(L^2 \cdot N_r^2)$ gram-matrix inner products; with $N_r = 20$ and $L = 12$, this is negligible relative to the main extraction pass. The ℓ_2 -normalization step costs $O(N \cdot M \cdot D)$ and the MMD² computation at each retained layer costs $O(N^2)$ kernel evaluations; both are small relative to feature extraction at $N = 500$. Total wall-clock time at $N = 500$ on a single NVIDIA A6000 GPU is approximately 7.87 s, measured with three read-out layers (one per canonical axis), compared to 80.92 s for Manifold Precision/Recall at the same sample size. Feature extraction requires a single forward pass through Φ regardless of M ; the additional layers contribute only hook-collection overhead, so wall-clock time scales sub-linearly with M .

Memory Embedding matrices for both sets at the M read-out layers require $2MND \times 4$ bytes in single-precision floating point. At the canonical three-axis configuration ($M = 3$ a-priori depths, $N = 500$, $D = 384$), this amounts to approximately 4.6 MB ($2 \times 3 \times 500 \times 384 \times 4$ bytes), well within the capacity of any contemporary evaluation GPU. Even the post-hoc CKA pass at its least aggressive setting ($\tau = 0.99$, $M = 11$ layers) requires only about 16 MB.

Hyperparameters Because the canonical metric fixes its read-out layers a priori (Section 3.3), it has *no* layer-selection or weighting hyperparameter. The fidelity axis uses a fixed, median-anchored five-bandwidth RBF ladder (Eq. (9)) and a permutation count B (default 500); the only deployment guidance is the sample size N (≥ 100 for monitoring, 500 for publication). The optional post-hoc CKA validation retains two legacy parameters, the redundancy threshold τ (default 0.80) and sub-batch count K (default 5), whose sensitivity is reported in Section 4.9.

3.8. Application to New Imaging Modalities

The M3-Score requires no retraining and no modality-specific configuration when applied to a *radiology* modality outside the BraTS brain MRI validation domain (Section 1.4). The procedure is as follows.

1. **Evaluation set preparation.** Assemble $N \geq 100$ real and generated images from the target modality. Images are resized to 224×224 and normalized to $[0, 1]$ by the RadioDINO preprocessing pipeline.
2. **A-priori layer read-out.** Extract CLS-token features at the three fixed layers $\ell_{\text{fid}} = 12$, $\ell_{\text{mem}} = 9$, $\ell_{\text{cov}} = 4$ (Section 3.3). No layer is selected adaptively, so the configuration is identical across modalities and reproducible without a calibration set.
3. **Axis computation.** Compute the fidelity MMD^2 at L12 (Algorithm 3) together with its permutation p -value, bootstrap 95% CI, and null-standardized separation Z ; the L9 memorization rate; and the L4 coverage precision/recall.
4. **Axis interpretation.** A large fidelity MMD^2 with significant Z indicates a distributional gap; a non-zero memorization rate flags near-duplicate leakage; low coverage indicates limited diversity (subject to the small- N caveat of Section 4.4).
5. **Per-image ranking.** Optionally execute Algorithm 4 to rank generated images by $s(\hat{x}_j)$ for downstream review.

For training-time monitoring, $N \geq 100$ images are sufficient to obtain stable estimates; for publication-grade evaluation, $N \geq 500$ is recommended. Increasing N reduces the variance of the fidelity $\widehat{\text{MMD}}^2$ at a cost proportional to the additional forward passes through Φ . The optional post-hoc CKA validation (Algorithm 5) may additionally be run on $N_r = 20$ real images to confirm that L12 remains the most discriminative fidelity layer for the new modality.

4. Results and Evaluation

We evaluate the M3-Score under one canonical configuration applied across every experiment: RadioDINO-s16 backbone, $N = 500$ where sample size permits, multi-bandwidth RBF kernel, and the unbiased U-statistic MMD^2 . The real reference set is $N = 500$ axial BraTS slices and the generated set is $N = 500$ images from an unconditional DDPM trained on the same distribution; representative samples of both sets are shown in Fig. 3. Real sets are drawn round-robin across subjects (Section 3.2); the canonical reference is $N = 500$ slices spanning 500 distinct subjects. The three axes read from the a-priori blocks L12, L9 and L4 (Section 3.3); no depth is selected by test outcome. The results cover the canonical three-axis table, a noise/blur sanity check, OOD and cross-modality discrimination (including the CMMD rank inversion), per-axis validation, coverage-estimator validation, a generation-quality ladder, pathology-region masking, sample efficiency, a post-hoc CKA ablation, and the permutation test. The Isolation Forest anomaly detector used in Section 4.3 is trained exclusively on real image embeddings and is completely independent of the M3-Score computation.

4.1. The Canonical Three-Axis Table

Table 2 reports all three axes for five comparison sets of increasing a-priori severity, measured against one reference under one configuration. The first row contains no generated images at all: it is a second block of real slices drawn from a *disjoint* pool of subjects by the same procedure as the reference, and it is the value the metric should return when nothing is wrong.

The null case reads as a null The held-out real probe registers $\widehat{\text{MMD}}^2 = 0.000$ with permutation $z = -0.6$, i.e. the fidelity axis correctly declines to distinguish two samples of real patients. This is the check that the reference protocol of Section 3.2 exists to pass, and it is not automatic: the same comparison against a reference drawn in filename order does *not* pass, because the two real blocks then straddle a site boundary rather than sampling one population. Coverage on this row is 0.970 and the memorization rate 0.478, both as expected: real slices do have real near-neighbours, and a real cohort does cover the manifold.

The axes order severity, and they are not redundant Fidelity increases monotonically across the five tiers ($0.000 < 0.073 < 0.216 < 0.398 < 0.459$; Spearman $\rho = 1.00$ against the a-priori severity rank) and coverage decreases monotonically. The two axes nonetheless carry different information, which is the point of reporting them

Table 2

The canonical three-axis table. One reference ($N = 500$ BraTS slices from 500 distinct subjects, drawn round-robin across subjects), one configuration (RadioDINO-s16, multi-bandwidth RBF, unbiased U -statistic), three a-priori depths. Comparison sets are ordered by a-priori severity. The first row contains only real images, drawn from a subject-disjoint pool. p is the one-sided permutation p -value at $B = 500$ shuffles, whose resolution floor is $1/501 = 0.002$; z is the null-standardized separation, reported as a magnitude only (Section 3.3). Bootstrap CI at 200 resamples. [‡]Not interpretable: degenerate U -statistic under H_0 .

Comparison set	Content	Fidelity (L12)				Memorization (L9)	Coverage (L4)
		$\widehat{\text{MMD}}^2$	95% CI	p	z	rate	Cov. $\uparrow$
Real probe	real vs. real	0.0000	[0.0010, 0.0026] [‡]	1.000	-0.6	0.478	0.970
DDPM	in-domain diffusion	0.0732	[0.0705, 0.0801]	0.002	342.4	0.010	0.384
WDM-3D (MRI)	weaker same-modality generator	0.2159	[0.2102, 0.2250]	0.002	696.9	0.000	0.002
LIDC CT	cross-modality shift	0.3984	[0.3909, 0.4117]	0.002	718.4	0.000	0.000
Retinal	far-domain shift	0.4585	[0.4525, 0.4678]	0.002	805.3	0.000	0.000

separately. The DDPM sits at a modest fidelity distance of 0.073 (nearer the real probe than any other generated set, and its samples are individually plausible) while covering only 38.4% of the real manifold. A single scalar cannot express that combination. The weaker same-modality generator separates the axes further still: its fidelity distance roughly triples relative to the DDPM, but its coverage collapses by a factor of nearly 200, to 0.002.

Memorization is near-zero for the DDPM The memorization rate at L9 is 0.010 for the DDPM against 0.478 for the real probe. The generator is not reproducing its training data; the axis registers what it should on the positive control (real slices, which genuinely do have near-duplicates) and near-nothing on the generator.

A caveat on the null row's interval The bootstrap interval on the real probe, [0.0010, 0.0026], does not contain its own point estimate of 0.0000. This is the expected behaviour of the ordinary bootstrap on a degenerate U -statistic (Arcones and Giné, 1992): under H_0 the MMD^2 estimator has a non-normal limiting distribution and the naive resampling interval is not consistent there. The interval is therefore not interpretable on the null row, and we report it struck through rather than silently omitting it. Intervals on the remaining rows, which are measured under the alternative, are unaffected.

What the permutation column does and does not show Every generated row is rejected at the resolution floor of the test, and the real probe is not rejected. That is the correct behaviour, and it is worth having: FID and KID report no such quantity at all. But the four generated rows are indistinguishable on p , and their z values, while ordered here, are magnitudes whose scale grows with N by construction. The permutation null is exact equality of the two distributions, a hypothesis that no usable generative model satisfies, so rejecting it is not evidence of poor quality, and failing to separate two generators on p is not evidence that they are comparable. We use the column as a sanity check on the null row and report it descriptively elsewhere. Section 5.1 returns to this.

Reported quantities The tables below use the following quantities throughout. ρ is the Spearman rank correlation between a perturbation level and a metric's value; a metric that increases monotonically with degradation attains $\rho = 1$. τ is Kendall's rank correlation, used where the ordering rather than the spacing of a small number of tiers is at issue. CV is the coefficient of variation, the standard deviation across repeated subsamples divided by their mean, expressed as a percentage; lower means a metric returns a more repeatable value at a given N . Z is the null-standardized separation defined in Section 3.3, reported as a magnitude of separation from the permutation null rather than as a Gaussian z -score. α -Precision is the fraction of generated samples that fall inside the k -NN manifold of the real set ($k = 5$), and β -Recall the fraction of real samples that fall inside the k -NN manifold of the generated set; the two measure fidelity and coverage respectively, and both lie in $[0, 1]$ with higher being better (Kynkäänniemi et al., 2019; Alaa et al., 2022). Δ denotes a paired difference between two rows of the same table, reported with its bootstrap 95% CI.

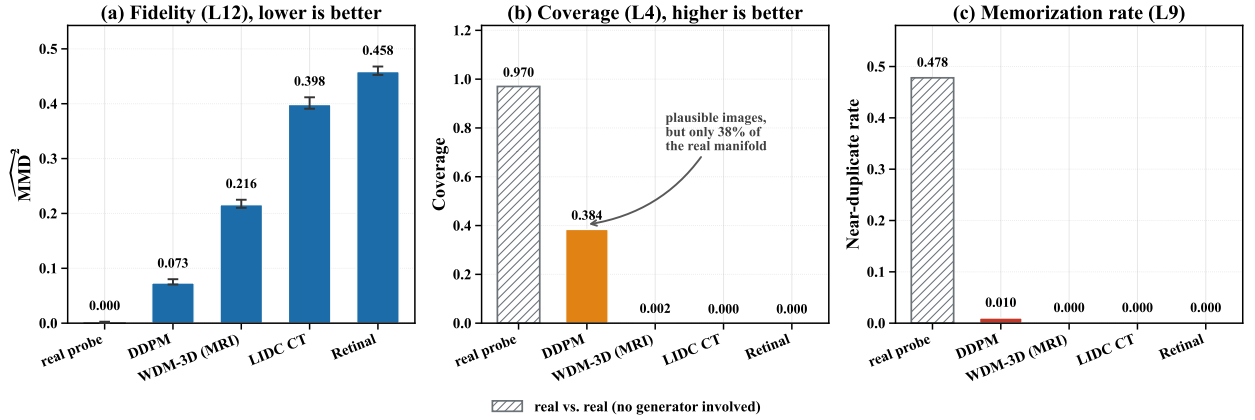


Figure 2: The three axes across five comparison sets of increasing a-priori severity, all against one reference of $N = 500$ BraTS slices from 500 distinct subjects. The leftmost bar in each panel is a real-vs-real probe drawn from a subject-disjoint pool, shown hatched because no generator is involved: it is the value each axis should return when nothing is wrong. (a) Fidelity orders the tiers exactly (error bars are bootstrap 95% intervals; the probe’s interval is omitted as it is not interpretable on a degenerate U -statistic). (b) Coverage separates what fidelity cannot: the DDPM lies closest to the real data of any generated set, yet populates barely a third of the manifold. (c) Memorization is near-zero for every generator and high for the real probe, whose slices genuinely do have real near-neighbours.

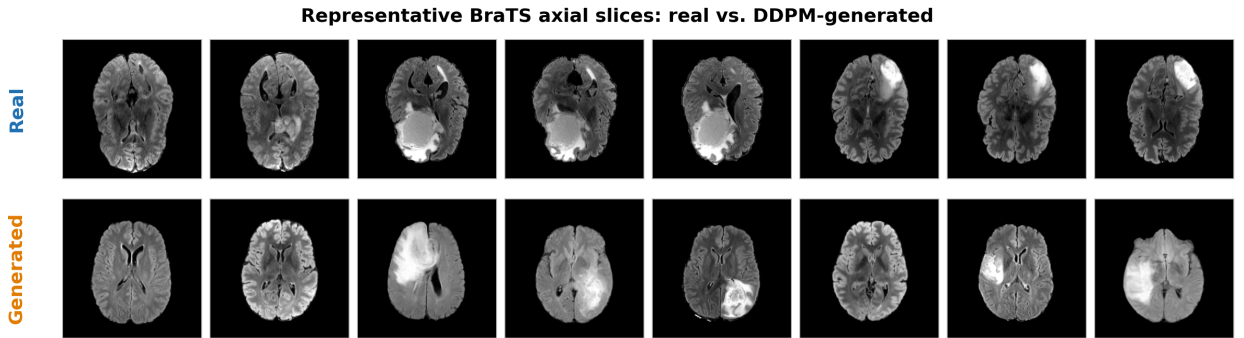


Figure 3: Representative axial BraTS slices: real (top) and unconditional DDPM-generated (bottom). Both sets exhibit plausible tissue boundaries, ventricles, and FLAIR-hyperintense lesions, and are difficult to separate by visual inspection, which motivates a rigorous distributional metric with native significance testing rather than a qualitative check. Images are shown at the 224×224 resolution fed to the encoder.

4.2. Baseline Sanity Check: Noise and Blur Robustness

A necessary condition for any evaluation metric is monotonic response to known image degradation. We inject progressive Gaussian noise ($\sigma \in \{0.0, 0.01, 0.02, 0.05, 0.1, 0.2, 0.5\}$) and Gaussian blur (radius $r \in \{0, 1, 2, 3, 4\}$ pixels) into the real image set and compute all nine metrics between the clean reference and the perturbed set. FRD is omitted from this particular sweep because it is computed from disk-resident image directories rather than in-memory perturbed tensors; it is reported on the primary comparison in the radiomic-feature literature (Konz et al., 2026). Table 3 reports Spearman rank correlations between perturbation level and metric value, together with 95% confidence intervals via Fisher z-transformation.

Noise degradation At $\sigma = 0$ (clean versus clean), M3 returns exactly 0.0, consistent with the unbiased $\widehat{MMD}^2$ estimator whose expectation is zero under identical distributions. All nine metrics achieve perfect Spearman correlation ($|\rho| = 1.0$, $p < 0.001$) under Gaussian noise, confirming that M3 responds correctly to progressive pixel-level

Table 3

Spearman rank correlation ρ between perturbation level and metric value ($N = 7$ noise levels; $N = 5$ blur radii). $\uparrow$: metric should increase with perturbation. $\downarrow$: metric should decrease. All $p < 0.001$ except α -Precision and β -Recall under blur ($p = 0.005$).

Metric	Dir.	Spearman ρ	
		Noise	Blur
M3-Score (ours)	$\uparrow$	1.000	1.000
FID (Inception)	$\uparrow$	1.000	1.000
KID (Inception)	$\uparrow$	1.000	1.000
SSIM	$\downarrow$	-1.000	-1.000
PSNR	$\downarrow$	-1.000	-1.000
MS-SSIM	$\downarrow$	-1.000	-1.000
LPIPS	$\uparrow$	1.000	1.000
α -Precision	$\downarrow$	-0.982	-0.975
β -Recall	$\downarrow$	-0.982	-0.975

degradation (Fig. 4 and Fig. 5). As noise increases to $\sigma = 0.5$, the fidelity MMD² rises to 0.607, FID to 521.96, and KID to 0.868, while SSIM collapses to 0.031 and PSNR to 9.29 dB (raw unnormalized values; Fig. 4 and Fig. 5 display all metrics normalized to [0, 1] for visual comparison).

Blur degradation Under Gaussian blur the M3-Score achieves $\rho = 1.000$ ($p < 0.001$), identical to FID and KID, and the sequence is strictly monotone across all five radii. The α -Precision metric reveals that blur is substantially more destructive to manifold membership than noise at comparable image quality. At blur radius $r = 2$ (PSNR = 27.88 dB), α -Precision collapses to 0.006: virtually no perturbed images remain within the real clinical manifold. At Gaussian noise $\sigma = 0.05$ (PSNR = 27.96 dB, similar quality), α -Precision remains 0.384. This confirms that the real BraTS image manifold is defined primarily by high-frequency texture structure that blur destroys while additive noise partially preserves.

4.3. Unsupervised Anomaly Detection and OOD Correlation

An earlier version of this section asked whether the M3 per-image score agrees with a label-free anomaly detector better than an InceptionV3 distance does. We report here why that question, posed that way, cannot be answered, and then answer the question it should have been.

A circular design, and the measurement that exposes it The detector is an Isolation Forest (Liu et al., 2008) fitted on embeddings of the real reference and used to score each generated image. The difficulty is that it must be fitted *in some feature space*, and that space is one of the spaces under comparison. A distance computed in the same space as the detector is then trivially advantaged.

To measure how large that advantage is, we fit the detector separately in each space and cross-tabulate (Table 4). The structure is unambiguous: each distance tracks the detector fitted in its own space ($\rho = +0.736$ for M3 under a RadioDINO-fitted detector, +0.649 for the InceptionV3 proxy under an Inception-fitted one) and barely tracks the other (+0.155 and +0.144 respectively). The diagonal exceeds the off-diagonal by a factor of four to five in both directions, and the confidence intervals are disjoint.

The experiment therefore measures agreement between two statistics computed in the same representation, not the ability to detect clinical anomaly. It ranks no backbone above another, and we draw no comparative conclusion from it. We report it because the design is a natural one, appears in this literature, and its circularity is not obvious until the symmetric version is run. Notably, pixel MSE, which belongs to neither space, correlates modestly and *about equally* with both detectors (+0.288, +0.364), which is what an uninvolved third measure should do.

The question that can be answered What a practitioner needs is not agreement with a fitted detector but discrimination against a ground truth that no feature space had a hand in choosing. Image provenance supplies one: an image is either a held-out real scan or it is not. The remainder of this section uses that criterion.

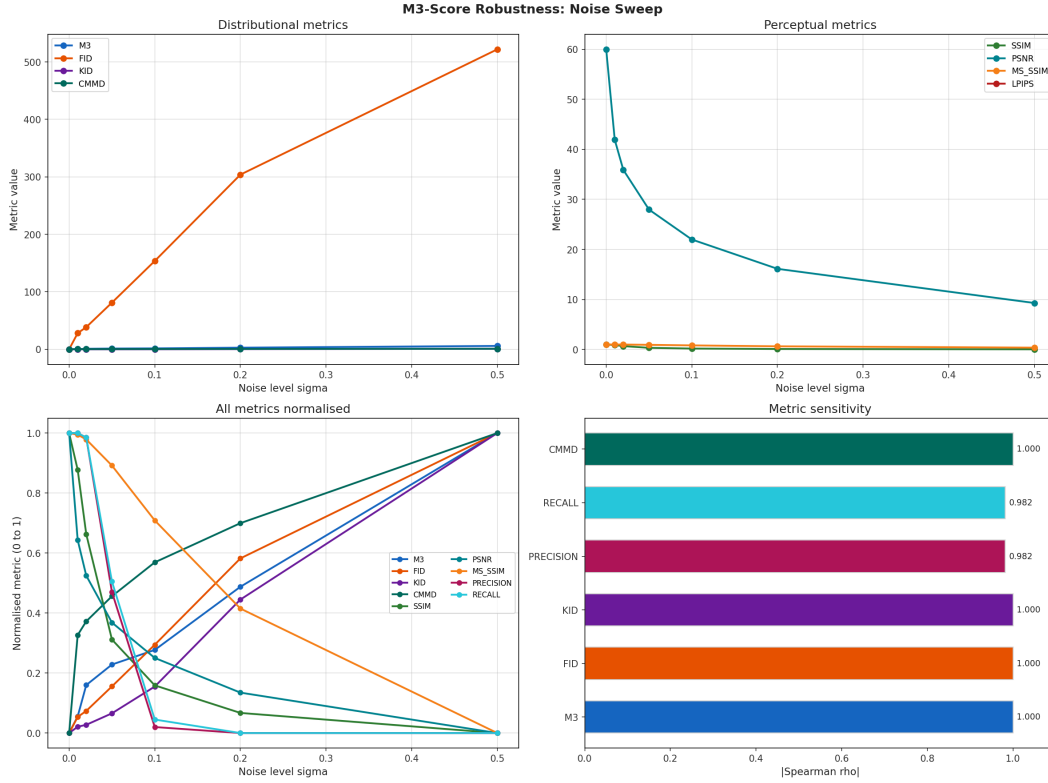


Figure 4: Gaussian noise sweep ($\sigma \in [0, 0.5]$, $N = 200$ BraTS images). The composite panel shows four views: (a) distributional metrics (M3/FID/KID) in raw units, (b) perceptual metrics (SSIM/PSNR/MS-SSIM/LPIPS), (c) all metrics normalized to $[0, 1]$ for visual comparison, and (d) $|\text{Spearman } \rho|$ bar chart. All nine metrics achieve $|\rho| = 1.00$ ($p < 0.001$); the normalized panel (c) shows M3 and SSIM/PSNR diverge earliest, indicating higher sensitivity at low perturbation levels. Note: raw metric values cited in text (e.g. fidelity $\text{MMD}^2 = 0.607$, $\text{FID} = 521.96$ at $\sigma = 0.5$) correspond to unnormalized panel (a)/(b); panel (c) shows the same data rescaled.

Table 4

Spearman correlation between a per-image distance and an Isolation Forest anomaly score, with the detector fitted in each feature space in turn ($N = 500$ generated images, 95% CI via Fisher z). Bold marks the cell where distance and detector share a representation. The diagonal dominates, so this design measures same-space agreement rather than anomaly-detection quality, and cannot be used to rank feature spaces.

Per-image distance	Isolation Forest fitted in	
	RadioDINO-s16	InceptionV3
M3-Score (RadioDINO, L2-norm)	+0.736 [+0.69, +0.77]	+0.144 [+0.06, +0.23]
FID proxy (InceptionV3)	+0.155 [+0.07, +0.24]	+0.649 [+0.60, +0.70]
Pixel MSE	+0.288 [+0.21, +0.37]	+0.364 [+0.29, +0.44]

Per-image separation by feature space To quantify whether radiology-pretrained or natural-image features better separate real from out-of-distribution images, we hold the scoring rule fixed and vary only the feature space. Each image is scored by the L2 distance from its embedding to the centroid of the real reference, on L2-normalised features so the three spaces are commensurable, and we compute ROC-AUC between a held-out real probe (subject-disjoint from the reference) and each OOD tier. Table 5 reports the result.

The comparison only exists on the hard tier Two of the four tiers are saturated: against cross-modality CT and against retinal fundus, every feature space scores ≥ 0.998 , so those tiers separate no backbone from any other and

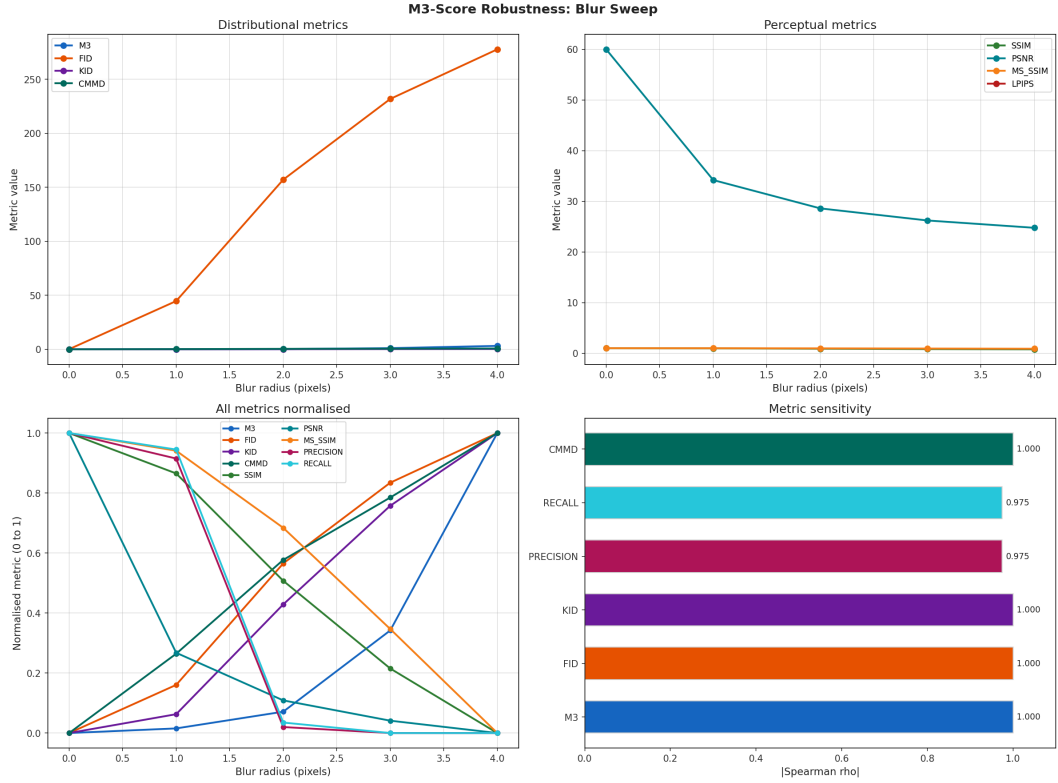


Figure 5: Gaussian blur sweep ($r \in \{0, 1, 2, 3, 4\}$ pixels, $N = 200$ BraTS images). The composite panel shows four views: (a) distributional metrics (M3/FID/KID) in raw units, (b) perceptual metrics (SSIM/PSNR/MS-SSIM/LPIPS), (c) all metrics normalized to $[0, 1]$ for visual comparison, and (d) |Spearman ρ | bar chart. M3-Score achieves $\rho = 1.000$ ($p < 0.001$). α -Precision collapses to 0.006 by $r = 2$, while distributional metrics remain monotonic, revealing that the real BraTS manifold is defined by high-frequency texture structure that Gaussian blur selectively destroys.

Table 5

Per-image OOD discrimination by feature space. ROC-AUC separating a held-out real probe (subject-disjoint) from each OOD tier, using one scoring rule (L2 distance to the real-reference centroid on L2-normalised features) in three feature spaces. $N = 500$ per set, reference of 500 slices from 500 distinct subjects. Tiers are reported separately rather than pooled because the two far tiers are saturated for every backbone and cannot rank them.

Feature space	Discriminative		Saturated	
	DDPM	WDM-3D	LIDC CT	Retinal
RadioDINO-s16 L12 (ours)	0.819	0.986	1.000	1.000
InceptionV3 2048-d (FID)	0.555	0.967	1.000	0.999
CLIP ViT-L/14 (CMMD)	0.582	0.999	0.998	0.999

we draw no conclusion from them. An earlier version of this experiment reported a single pooled AUC dominated by exactly those saturated tiers; reporting per tier shows where the backbones actually differ.

They differ on the tier that matters. Separating real brain MRI from *in-domain* DDPM output is the case a practitioner faces when deciding whether synthetic data is usable, and it is the only tier on which the three spaces are far apart. RadioDINO-s16 reaches ROC-AUC = 0.819, while InceptionV3 (0.555) and CLIP (0.582) sit close to chance: measured as margin over chance, the radiology backbone is roughly 5.8 $\times$ and 4.5 $\times$ better than the two natural-image backbones that underlie FID and CMMD. A metric built on either of those spaces cannot reliably tell a real brain MRI from this generator’s output at the per-image level; one built on RadioDINO can, imperfectly.

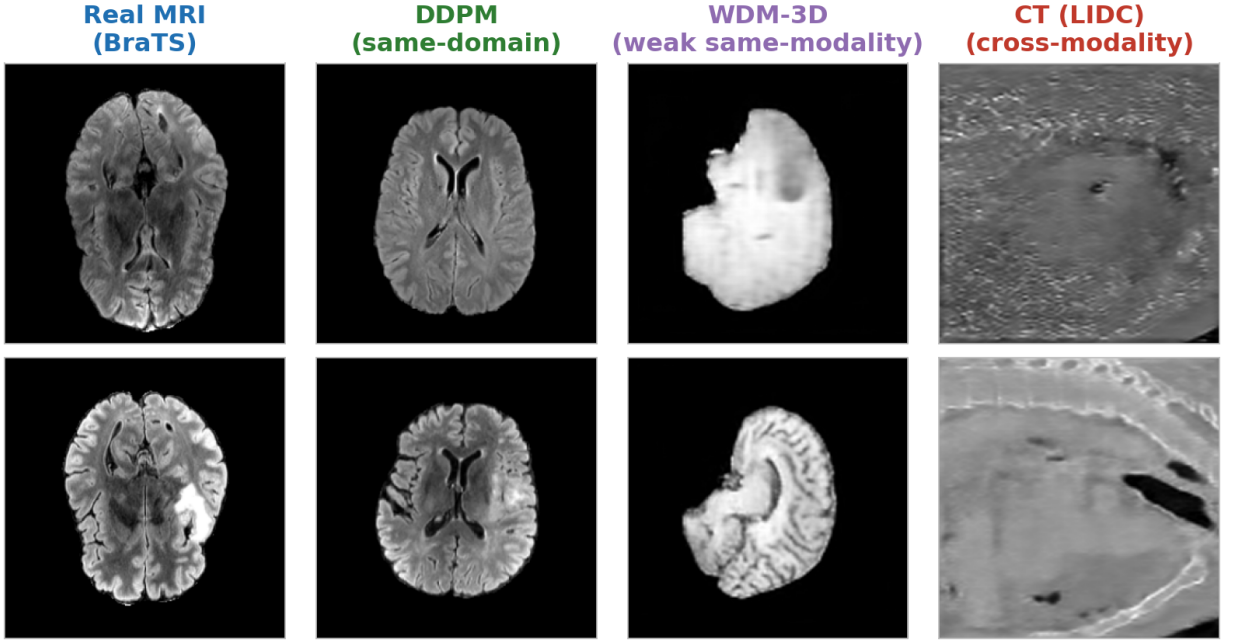


Figure 6: The three tiers evaluated against real BraTS MRI. From left: real MRI; a same-domain DDPM generator; a weaker same-modality WDM-3D generator (visibly lower-quality but still brain MRI); and a cross-modality LIDC chest CT scan, which is a different modality and anatomy. A faithful metric should rank the CT as the most distant from real brain MRI. M3 and FID do; CMMD inverts the last two, ranking the CT (right) as closer to real MRI than the weak WDM-3D brain generator (Table 6).

We report one result against ourselves: on the weak same-modality generator, CLIP (0.999) edges out RadioDINO-s16 (0.986). The radiology backbone is not uniformly superior, and we do not claim it is. Its advantage is concentrated where the discrimination is hard and in-domain.

An observed CMMD rank inversion on medical OOD ordering The most consequential comparison is not the aggregate AUC but a specific ordering failure. We evaluate three tiers against BraTS real ($N = 500$ slices from 500 distinct subjects, drawn as in Section 3.2). Bootstrap 95% intervals resample over *clusters* rather than rows: subjects on the real side, images on the generated side. Resampling real slices independently would treat near-duplicate slices of one patient as independent observations and yield intervals that are too narrow, which is the error the reference protocol exists to avoid. The tiers are a same-domain DDPM generator, a weaker same-domain (WDM-3D) generator, and a full cross-modality shift to chest/lung CT (LIDC). CMMD here follows the standard protocol of Jayasumana et al. (2024): OpenAI CLIP ViT-L/14 image features and an unbiased Gaussian-RBF MMD² with the median-heuristic bandwidth computed on the pooled real-plus-generated sample, so the inversion below cannot be attributed to a non-standard estimator or bandwidth. The correct order is DDPM < WDM-3D < CT, since a different modality should be the most distant; the three tiers are illustrated in Fig. 6. The M3 fidelity axis recovers this order with non-overlapping CIs ($0.073 < 0.216 < 0.398$; Table 6). CMMD, however, *inverts* the last two tiers: it assigns the cross-modality CT (0.664 [0.652, 0.682]) a *smaller* distance than the weak same-modality MRI generator (0.737 [0.719, 0.754]), ranking lung CT as closer to brain MRI than a poor brain-MRI generator. The two intervals are disjoint, so this is a rank inversion rather than sampling noise; the difference is $\Delta_{\text{CT-MRI}} = -0.073$, against $+0.183$ for M3. FID (InceptionV3) also orders the tiers correctly ($68.4 < 173.4 < 319.9$), so the failure is specific to *CLIP* features rather than to natural-image backbones in general: CLIP does not encode the radiology modality boundary that a radiology-trained backbone, and even InceptionV3, does.

Table 6

Cross-tier separation from BraTS real ($N = 500$; bootstrap 95% CIs for M3 and CMMD over 500 resamples). Higher = more distant, so the correct order is DDPM < WDM-3D < CT. M3 and FID order the tiers correctly; CMMD *inverts* the last two, ranking cross-modality CT as closer than the weak MRI generator ($\Delta_{\text{CT-MRI}} < 0$, CI excludes zero). FID values are point estimates (InceptionV3).

Comparison vs. BraTS real	M3 (fid.) [95% CI]	FID	CMMD [95% CI]
DDPM (same-domain)	0.073 [0.070, 0.079]	68.4	0.104 [0.102, 0.111]
WDM-3D (weak same-domain)	0.216 [0.211, 0.225]	173.4	0.737 [0.719, 0.754]
LIDC CT (cross-modality OOD)	0.398 [0.390, 0.412]	319.9	0.664 [0.652, 0.682]
CT – WDM-3D Δ	+0.183	+146.5	−0.073

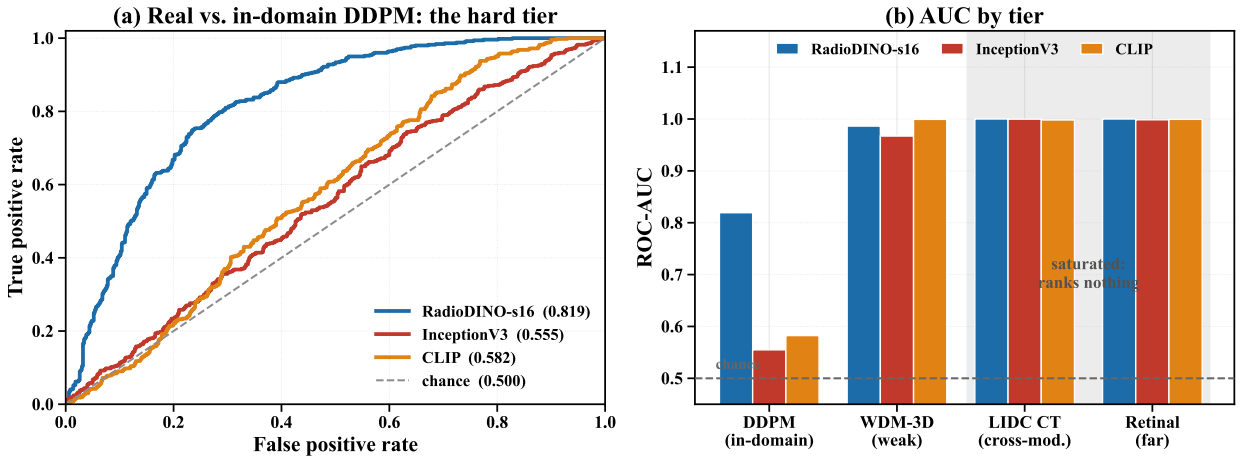


Figure 7: Per-image OOD discrimination, holding the scoring rule fixed and varying only the feature space. Each image is scored by its L2 distance to the centroid of the real reference; the positive class is a held-out real probe drawn from a subject-disjoint pool. **(a)** ROC on the only tier where the three spaces differ, real versus in-domain DDPM output. RadioDINO-s16 reaches AUC = 0.819; the InceptionV3 space underlying FID (0.555) and the CLIP space underlying CMMD (0.582) sit near chance, so a metric built on either cannot reliably tell a real brain MRI from this generator’s output at the per-image level. **(b)** AUC by tier. The two far tiers (shaded) are saturated at ≥ 0.998 for every space and rank nothing; a single pooled AUC, as reported in earlier work including our own, is dominated by exactly those uninformative tiers.

Domain-ontology ordering: a scope limitation Across five test sets ordered by intuitive OOD severity, FID and KID achieve Kendall $\tau = 1.000$ while M3 achieves $\tau = 0.800$ ($p = 0.083$). The single inversion is that M3 ranks retinal fundus images as *further* from brain MRI than chest CT. This is not an error: under a radiology-domain ontology, CT and MRI are neighbors and fundus is out-of-domain, which RadioDINO correctly reflects. It does, however, mean M3 should not be read as a universal visual-distance metric, and we restrict its claims to the radiology domain accordingly.

4.4. Per-Axis Validation

We validate the memorization and coverage axes by injecting the specific failure each is designed to detect and checking that the corresponding axis, and only that axis, responds. Features are read from the a-priori layers $\ell_{\text{mem}} = 9$ and $\ell_{\text{cov}} = 4$ (Section 3.3); $N = 150$ BraTS real and DDPM-generated images.

Memorization axis (validated) We replace a known fraction of generated images with near-duplicates of real images (Gaussian jitter, $\sigma = 5 \times 10^{-4}$) at injection rates $\{0.0, 0.1, 0.2, 0.3, 0.5\}$. The L9 memorization rate tracks the injected fraction to within a mean absolute error of 0.008 (0.010, 0.108, 0.210, 0.308, 0.506; Table 7), confirming the axis flags near-duplicates at the intended depth. The residual 0.010 at zero injection is the generator’s own near-duplicate rate, matching Table 2. Fidelity falls over the same ladder (from 0.073 to 0.017) and coverage rises (from 0.384 to 0.982),

Table 7

Per-axis injection test on the canonical reference ($N = 500$ from 500 subjects). A known fraction of the generated set is replaced by jittered near-duplicates of real images. The memorization rate recovers the injected fraction (mean absolute error 0.008); fidelity falls and coverage rises, because injecting real images genuinely moves the generated set toward the real distribution. The three axes are therefore correlated under this particular perturbation rather than independent, which is expected: it is a perturbation that changes all three underlying properties at once.

Injection rate	Mem. rate (L9)	Fidelity MMD ² (L12)	Coverage (L4)
0.0	0.010	0.0732	0.384
0.1	0.108	0.0597	0.674
0.2	0.210	0.0470	0.810
0.3	0.308	0.0346	0.906
0.5	0.506	0.0173	0.982

because injecting genuine real images moves the generated set toward the real distribution on every axis at once; under this perturbation the three are correlated rather than orthogonal.

4.5. Coverage Axis: Estimator Validation and a Negative Control

The coverage axis must fall when generated diversity is removed. We test this directly with a mode-drop ladder: a single random ordering of the generated set is drawn and each level retains a prefix of it, so a sample removed at one level stays removed at every deeper level. The ladder is *nested* by design, because a generator that loses modes does not redraw its output; sampling each level independently would confound removal with resampling noise. Drop fractions are $\{0, 0.2, 0.4, 0.6, 0.8\}$, $N = 500$ real slices from 500 distinct subjects, $k = 5$. We fixed the acceptance criterion before the run: at the a-priori depth L4, Spearman $\rho(\text{drop}, \cdot) \leq -0.9$ with a dynamic range of at least 0.05 across the ladder. The range condition matters because a statistic that is monotone but pinned near 0 or 1 passes the correlation test while carrying no usable signal.

We run the ladder against both estimators of Section 3.3, at all twelve blocks (Table 8, Fig. 8).

The recall estimator fails everywhere, in the wrong direction k -NN manifold recall rises as diversity is removed at *every* one of the twelve blocks, with ρ between $+0.70$ and $+1.00$. At the a-priori depth L4 recall climbs from 0.086 to 0.308 while 80% of the generated set is discarded; at L9 it climbs from 0.008 to 0.148. That the sign is wrong at all twelve depths is itself the diagnosis. Encoder blocks differ substantially in what they represent, so a defect rooted in the representation would vary with depth. A defect that is uniform across depth is a property of the estimator, which does not change with depth: here, the radius inflation described in Section 3.3.

The real-radius estimator passes everywhere Coverage as defined in Eq. (2) falls monotonically at all twelve blocks, $\rho = -1.00$ throughout, exactly as the by-construction argument requires. At the a-priori L4 it declines $0.384 \rightarrow 0.360 \rightarrow 0.312 \rightarrow 0.260 \rightarrow 0.152$ across the ladder, a dynamic range of 0.232. The a-priori depth therefore satisfies the pre-registered criterion, and does so comfortably.

Depth does matter for the range rather than the direction. The widest dynamic ranges sit at the shallow blocks (0.376 at L1, 0.232 at L4) and compress with depth (0.038 at L10 and L11), so the two deepest-but-one blocks are monotone yet too compressed to be useful, failing the range condition. This is consistent with the structural reason L4 was chosen a priori: coverage is a question about spatial and structural diversity, which is encoded early and progressively discarded as the representation becomes semantic. We note the agreement, but it is a post-hoc observation and we do not offer it as evidence that L4 is optimal.

Why we report the failed estimator The recall result is retained in Table 8 as a negative control. Its value is that it establishes the check has discriminating power: a sanity test that every candidate passes tells the reader nothing about the one that was adopted. It is also a caution that transfers beyond this paper. k -NN precision and recall are used widely as generative diagnostics, and their radii are estimated on the sets being compared, so any protocol that subsamples one of those sets, mode-drop probes and sample-size sweeps included, can invert the statistic's direction while appearing to measure diversity.

Table 8

Mode-drop response of the two coverage estimators at all twelve encoder blocks. ρ is Spearman correlation between drop fraction and the statistic; a usable coverage axis requires ρ strongly *negative*. Range is max–min across the ladder. The a-priori depth L4 is shown in bold. $N = 500$ real slices from 500 subjects, $k = 5$, nested ladder.

Block	k -NN recall (Kynkäänniemi et al., 2019)	Coverage (Naeem et al., 2020a), Eq. (2)		Verdict
	$\rho(\text{drop, recall})$	$\rho(\text{drop, cov.})$	range	
L1	+1.00	−1.00	0.376	pass
L2	+0.70	−1.00	0.196	pass
L3	+0.90	−1.00	0.228	pass
L4	+0.90	−1.00	0.232	pass
L5	+1.00	−1.00	0.192	pass
L6	+1.00	−1.00	0.144	pass
L7	+0.70	−1.00	0.108	pass
L8	+0.90	−1.00	0.086	pass
L9	+1.00	−1.00	0.060	pass
L10	+1.00	−1.00	0.038	range
L11	+1.00	−1.00	0.038	range
L12	+1.00	−1.00	0.050	pass

Recall: 0/12 blocks correct in direction. Coverage: 12/12 correct, 10/12 also clearing the range floor.

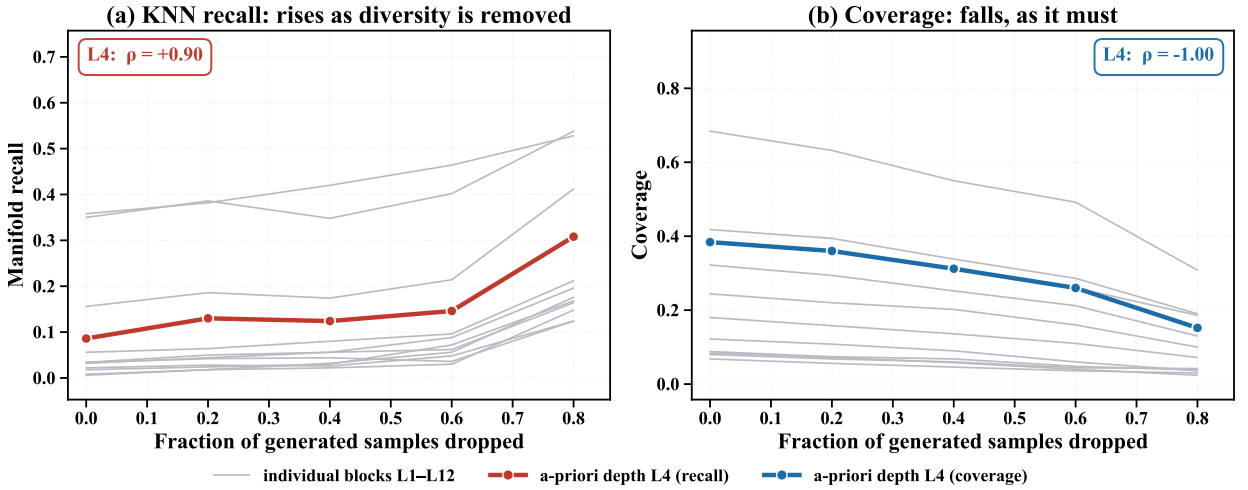


Figure 8: Mode-drop response at all twelve encoder blocks (grey), with the a-priori depth L4 highlighted. **Left:** k -NN manifold recall rises as generated diversity is removed, at every depth: its hyperspheres are fitted on the set being subsampled, so they widen as it thins. **Right:** the real-radius Coverage estimator falls monotonically at every depth, as its construction requires. A usable coverage axis must slope downward.

4.6. On the Choice of Depth

Fixing the three depths a priori protects against outcome-driven selection, but it invites the question of whether they are the right ones. We report what our own validations do and do not settle, and claim no more.

Memorization Under the canonical near-duplicate protocol of Section 4.4 (jitter $\sigma = 5 \times 10^{-4}$), the memorization rate recovers the injected fraction exactly at *all twelve* blocks: mean absolute recovery error is 0.000 everywhere. The test is saturated. That certifies the L9 prior costs nothing (no memorization result in this paper would change at another depth) while providing no evidence that L9 is preferable to any other block.

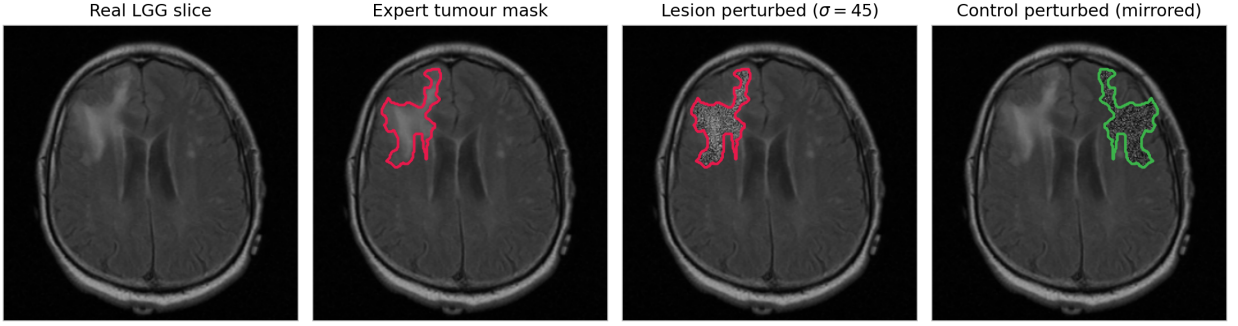


Figure 9: The area-matched lesion-specificity test on a real LGG slice. From left: the raw slice; the expert FLAIR-abnormality mask (red); the *LESION* condition (region-local Gaussian noise, $\sigma = 45$, inside the tumor, red); and the *CONTROL* condition (the same noise inside the mirrored, equal-area contralateral region, green). Lesion and control regions are identical in shape and area, so any difference in metric response is attributable to *where* the perturbation is applied, not to how much of the image is changed.

Making the test discriminative requires harder near-duplicates, and here we report a caution rather than a result. Under additive pixel jitter a detection ceiling appears and retreats toward the input as σ grows, which appears to favour shallow blocks. That ordering is not stable: it depends on the within-real nearest-neighbour threshold θ , and hence on how many distinct subjects the real reference contains, because a reference concentrated on few patients contains near-duplicate slices that depress θ directly. Depth conclusions drawn from a near-duplicate test are therefore entangled with reference composition, and we do not draw one. We report L9 as a structural prior that the canonical protocol shows to be harmless, and leave its optimality open.

Coverage Section 4.5 measures the mode-drop response at every block. Direction is depth-independent: the recall estimator is wrong at all twelve, the real-radius estimator correct at all twelve. Depth affects only dynamic range, which is widest at the shallow blocks where the a-priori L4 sits. No depth repairs the recall estimator, and no depth is needed to repair coverage.

Fidelity The L12 prior is the one choice with independent corroboration: the CKA analysis of Section 4.9 identifies it as the most discriminative non-redundant block. We note that this is a correlational check run on the same encoder rather than a held-out optimality argument.

A proper answer would measure every depth against every task on a reference whose composition is itself controlled, and would treat the resulting profile, rather than any single named block, as the object of interest. That study is larger than this paper and is the subject of follow-up work.

4.7. Lesion-Specificity with Real Expert Masks

A radiology-trained backbone should, in principle, respond more strongly to degradation of the diagnostically relevant region than to equal-area degradation of healthy tissue, a property natural-image backbones need not possess. We test this directly using the *real* expert FLAIR-abnormality segmentations of the lower-grade-glioma (LGG) dataset of Buda et al. (2019) (600 image-mask pairs), replacing the intensity-threshold proxy masks of earlier drafts.

The test is a within-image, area-matched contrast. For each image we perturb (i) the annotated tumor region [*LESION*] versus (ii) an equal-area, equal-shape region relocated to healthy tissue [*CONTROL*]. We use two independent control constructions: a *mirror* control (the tumor mask reflected to the contralateral hemisphere) and a *texture-matched* control (the tumor mask translated to the healthy-brain location whose local gradient energy best matches the tumor’s). Cases where a control lands off brain or on tumor are discarded, so lesion and control areas are identical by construction. We report the specificity ratio $\rho = \Delta_{\text{LESION}} / \Delta_{\text{CONTROL}}$, where Δ is the distributional shift (real vs. perturbed) induced by the perturbation; $\rho > 1$ means the metric reacts more to tumor damage than to equal-area healthy damage. The baseline is KID (unbiased Inception-MMD), which is valid at small N and lies in the *same estimator family* as M3, so the backbone is the only difference between the two. Figure 9 illustrates the two conditions.

Table 9

Lesion/healthy specificity ratio ρ with bootstrap 95% CIs, mirror control, $N = 200$. The kernel-controlled baseline (RBF on InceptionV3) differs from M3 *only* in the backbone; KID additionally differs in the kernel. Both Inception baselines behave alike, so the separation from M3 is backbone-driven, not kernel-driven. “n/a” marks σ below a metric’s detection floor.

σ	M3 (RadioDINO, RBF)	Inception, RBF	Inception, poly. (KID)
20	n/a	1.13 [0.85, 1.32]	1.12 [0.85, 1.34]
30	n/a	1.09 [0.88, 1.26]	1.08 [0.88, 1.25]
45	1.71 [0.94, 2.14]	1.10 [0.92, 1.26]	1.08 [0.92, 1.24]
60	1.57 [1.03, 1.97]	1.10 [0.95, 1.24]	1.08 [0.93, 1.22]
90	1.39 [1.06, 1.71]	1.08 [0.96, 1.20]	1.06 [0.93, 1.19]

Two regimes emerge. Under *gross erasure* (zeroing the region) no metric is lesion-specific ($\rho \approx 1.1$ – 1.3 for all): the black-hole signal dominates and swamps any semantic difference, consistent with, and explaining, the artifact-invariance of M3 discussed below. Under *subtle* region-local Gaussian noise, however, M3 responds more strongly when the perturbation falls inside the tumor than when the same perturbation falls in healthy tissue, whereas the Inception baselines respond almost equally to both. Figure 10 traces ρ against noise strength σ : in the measurable regime ($\sigma \geq 45$; below this the L12 MMD² underflows to zero at $N = 200$ and the ratio is undefined) M3 attains $\rho = 1.4$ – 1.8 against *both* controls, which agree to within 0.1, whereas the Inception baselines stay flat at $\rho \approx 1.03$ – 1.10 .

Isolating the backbone from the kernel Attributing this gap to the *backbone* requires holding the estimator fixed, since M3 uses a multi-bandwidth RBF kernel while KID uses a degree-3 polynomial one; a comparison against KID alone varies both. We therefore add a kernel-controlled baseline: the *identical* multi-bandwidth RBF MMD² of Algorithm 3, applied to InceptionV3 features instead of RadioDINO ones, so that the backbone is the only difference from M3. It behaves like KID, not like M3 (Table 9): across $\sigma \in \{45, 60, 90\}$ the RBF-on-Inception ratio is 1.03–1.10 and the polynomial-kernel KID ratio is 1.03–1.08, agreeing to within 0.04 at every σ , while M3 attains 1.39–1.81. Exchanging the kernel therefore accounts for essentially none of the gap, and the selectivity is attributable to the radiology backbone rather than to the estimator, to pixel magnitude, or to texture complexity; the last two are already controlled by the area-matched design and the 4% gradient-energy match of the mirror control.

Two caveats bound the claim. The effect decays as σ grows (extrapolating toward the $\rho \approx 1.1$ of gross erasure), so it is scoped as selectivity to *subtle* diagnostic-region degradation, not a universal pathology-sensitivity advantage. And at $\sigma = 45$ the M3 bootstrap interval still admits $\rho = 1$ ([0.94, 2.14] mirror); only at $\sigma = 60$ and 90 does the lower bound exceed unity (1.03 and 1.06 mirror; 1.02 and 1.03 texture-matched), so the individually significant evidence sits in that upper band rather than across the whole sweep. As no LGG generator was available, this is a metric-sensitivity test on real images under controlled perturbations, not a generator-quality result.

A second experiment isolates what each metric is actually responding to when a region is masked. Across three conditions ($N = 500$; Table 10) with a 64×64 center square mask on BraTS, masking the center of real images (condition B) moves them *closer* to the tumour-free generated distribution in RadioDINO space (the M3 fidelity axis falls by 39.9%), whereas FID *increases* by 211.0% and CMMD by 221.0%.

The direction of these changes identifies the cause. If FID and CMMD were responding to the removal of the tumor, masking it in real images would move those images *toward* the tumor-free generated set and the distances would fall, as M3’s does. They instead rise sharply, and they rise by a similar amount in condition C, where the same square is applied to generated images that contain no tumour to remove (+264.2% FID, +358.7% CMMD). Both metrics therefore respond to the black square itself rather than to the anatomical change, and the magnitude of the increase is not evidence of pathology sensitivity. M3 is comparatively invariant to the artifact and moves in the direction the anatomy implies. Read together with the LGG result, this is a statement about *specificity*: RadioDINO features track clinically meaningful change rather than the pixel-level artifacts that inflate FID and CMMD. The BraTS masks here are intensity-threshold FLAIR proxies, not ground truth, so this is reported as a supporting result only; the LGG experiment above, which uses real expert masks, carries the primary claim.

4.8. Sample Efficiency and Stability

Two different questions are usually conflated under “stability”. The first is *repeatability*: run the same evaluation twice at the same N , on fresh draws, and how much does the answer move? The second is *comparability*: run the

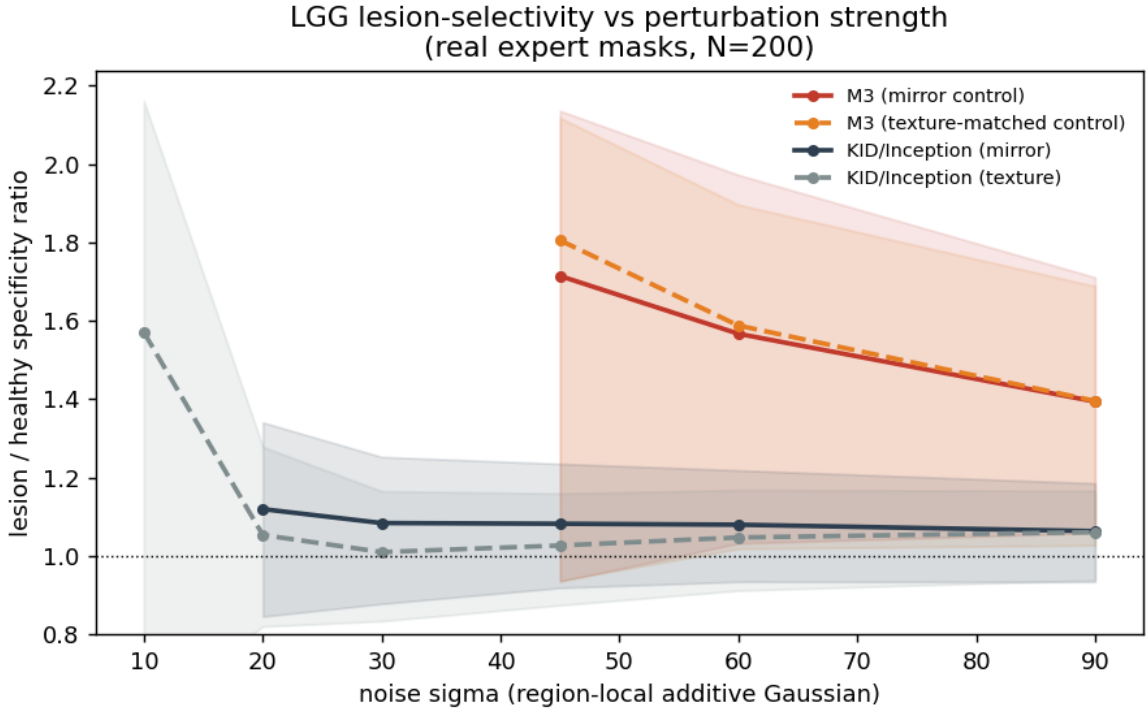


Figure 10: Lesion/healthy specificity ratio ρ versus region-local noise σ on real LGG expert masks ($N = 200$). M3 (red/orange) is lesion-selective ($\rho = 1.4$ – 1.8) against both the mirror and texture-matched controls, which nearly coincide; the Inception-MMD baseline (gray) stays at $\rho \approx 1.05$. Bands are bootstrap 95% CIs. Points below the M3 detection floor ($\sigma < 45$) are omitted.

Table 10

Masking with a 64×64 zero-filled center square, on the canonical reference ($N = 500$ from 500 subjects). B masks the center of real images; C masks the same region in the (tumour-free) generated images. Relative change is against the condition-A baseline, which matches Table 2. FID and CMMD rise by more than 200% in *both* B and C (including C, where no tumour is removed), which identifies the black square rather than the anatomical change as what they are responding to.

Condition	M3	FID	CMMD
A: real vs. gen (baseline)	0.0732	68.4	0.1040
B: real vs. masked-real	0.0440	212.7	0.3339
C: real vs. masked-gen	0.1192	249.0	0.4772
Rel. change (B–A)/A	–39.9%	+211.0%	+221.0%
Rel. change (C–A)/A	+63.0%	+264.2%	+358.7%

evaluation at two different N on the same underlying data, and does the metric mean the same thing? We measure both, because the metrics separate cleanly on them and in opposite directions.

Each real draw resamples *subjects* and then slices from them, so the variation reported is the variation a practitioner sees when assembling a reference, not the smaller variation obtained by reshuffling slices within a fixed subject pool. Ten draws per sample size.

Repeatability: FID is better below $N = 500$, and we say so Table 11 reports the coefficient of variation. FID is the more repeatable statistic at every sample size below 500: CV 3.6% against M3’s 6.7% at $N = 100$, and 9.0% against 13.0% at $N = 25$. The unbiased MMD² estimator underlying both M3 and CMMD carries higher small-sample

Table 11

Stability by sample size, ten independent draws each; the real side resamples subjects, not slices. **CV** is repeatability at fixed N (lower is better; best per row in bold). **Mean** is the value each metric returns: constant for the two unbiased U -statistics, drifting by 2.5 $\times$ for FID over the same range, on identical data.

N	CV (%), repeatability			Mean, comparability		
	M3	FID	CMMD	M3	FID	CMMD
25	12.99	8.98	11.97	0.0734	169.5	0.1019
50	9.71	4.32	6.57	0.0717	132.6	0.1003
100	6.73	3.63	7.44	0.0750	112.0	0.1038
200	5.60	2.25	2.10	0.0722	88.2	0.1001
500	1.27	1.25	1.17	0.0732	67.3	0.1028
drift (max/min of the mean)				1.05$\times$	2.52 $\times$	1.04$\times$

variance than FID’s Gaussian-moment estimator, and no framing of ours changes that. By $N = 500$ the three converge (1.27%, 1.25%, 1.17%). We therefore recommend M3 at $N \geq 100$ and preferably $N = 500$.

Comparability: FID’s value is not portable across N , M3’s is The mean tells a different story from the variance. Across a twentyfold change in sample size, on the same real pool and the same generator, the M3 fidelity value is essentially constant: 0.0717 to 0.0750, a spread of 4.4% and a max/min ratio of 1.05 $\times$. FID over the same range falls from 169.5 to 67.3, a ratio of **2.52 $\times$** . Nothing about the generator changed; only the number of images did.

This is the finite-sample bias of the Fréchet estimator, documented for natural images by Chong and Forsyth (Chong and Forsyth, 2020), reproduced here on radiology. Its practical consequence is that two studies reporting FID against different reference sizes are not comparing the same quantity, and the difference between them can exceed the difference between a good and a poor generator. An unbiased U -statistic has no such dependence, which is why M3 and CMMD are flat. CMMD drifts 1.04 $\times$, confirming the property belongs to the estimator rather than to our backbone.

The two paragraphs together give the honest recommendation. If a study is constrained to a small evaluation set and only ever compares within itself, FID is the more repeatable choice. If values are to be compared across studies, across reference sizes, or against a published number, FID cannot support the comparison and M3 can.

4.9. CKA Threshold Ablation

This is a *post-hoc check*, not a layer-selection mechanism: the canonical fidelity axis is fixed at L12 a priori (Section 3.3). We include it to ask whether an adaptive CKA procedure, given a free choice, would have preferred a different block. For each redundancy threshold τ , greedy backward CKA pruning is run on the real reference and the surviving layer set is recorded (Table 12).

What the check does establish **L12 survives at every threshold**, from the most aggressive pruning ($\tau = 0.70$, three layers retained) to no pruning at all ($\tau = 0.99$). No other block has that property: L1 drops out at $\tau = 0.70$, and the mid-depth blocks enter and leave as the cutoff moves. The a-priori fidelity depth is therefore the one representation an adaptive procedure never discards, which is the limited claim we make for it.

What it does not establish, and why the figure of merit changed Earlier work, including our own, summarised this sweep with a discriminability ratio $(S_{RG} - S_{RR})/S_{RR}$, where S_{RR} is the mean discrepancy between two real blocks. That ratio is not usable under a correct reference protocol. When the two real blocks are drawn across subjects rather than from one site, they are nearly indistinguishable ($S_{RR} \approx 2\text{--}12 \times 10^{-5}$, Table 12), so the ratio divides by a near-zero denominator and becomes both enormous and erratic: it ranges from 787 to 4176 across thresholds and is non-monotone in τ . Its apparent stability in earlier reporting (≈ 9 across all thresholds) was an artifact of a reference whose real-versus-real distance was inflated roughly a hundredfold. We therefore report S_{RG} and S_{RR} separately and treat the ratio as uninformative rather than as a figure of merit. S_{RG} itself is notably flat across thresholds (0.087–0.092), which says the extra layers admitted at looser cutoffs add little.

Across every threshold the most aggressive pruning ($\tau = 0.70$) retains exactly {12}, and the deep block L12 is never dropped. The discriminability curve is non-monotone in τ , but the qualitative conclusion is invariant: separability

Table 12

CKA threshold ablation on the canonical reference ($N = 500$ from 500 subjects). S_{RG} is the mean MMD^2 between real and generated over the active layers; S_{RR} the same between two subject-disjoint real blocks. L12 (bold) is retained at every threshold. The discriminability ratio is shown for continuity with earlier reporting but is not interpretable here: S_{RR} is near zero by design under a correct reference, so the ratio is dominated by its denominator.

τ	M	Active layers	S_{RG}	S_{RR}
0.70	3	{2, 5, 12 }	0.0915	1.2×10^{-4}
0.80	4	{1, 2, 6, 12 }	0.0866	0.6×10^{-4}
0.85	5	{1, 2, 4, 7, 12 }	0.0907	0.2×10^{-4}
0.90	6	{1, 2, 4, 6, 8, 12 }	0.0879	0.4×10^{-4}
0.92	6	{1, 2, 4, 6, 8, 12 }	0.0879	0.4×10^{-4}
0.95	7	{1, 2, 4, 5, 7, 9, 12 }	0.0900	0.7×10^{-4}
0.97	8	{1, 2, 4–7, 9, 12 }	0.0890	0.9×10^{-4}
0.99	12	{1– 12 }	0.0874	0.6×10^{-4}

is carried by the deepest block, which is precisely the a-priori fidelity layer. The most aggressive pruning we apply retains three blocks and L12 is among them; no threshold discards it. This ablation does not feed back into the metric definition, and it does not show L12 to be optimal, only that it is the block a redundancy criterion never removes, which is the weaker claim we rely on (Section 4.9).

4.10. Statistical Significance: Permutation Test

We construct a null distribution by pooling the real and generated L12 fidelity features and permuting the labels 500 times, recomputing the unbiased MMD^2 on each shuffle (Section 3.3). This is the native fidelity-axis permutation test, not an auxiliary construction.

The observed fidelity MMD^2 for the DDPM is 0.0732. No permuted statistic out of $B = 500$ reached it. Using the standard add-one estimator, which avoids reporting an impossible $p = 0$, this gives $\hat{p} = (0 + 1)/(B + 1) = 1/501 = 0.002$. This is the smallest value the test can return at $B = 500$: it is a resolution floor, not evidence of a smaller p , and we therefore write $\hat{p} = 0.002$ rather than $p < 0.002$. A smaller reportable p would require more permutations, not a larger effect.

We do *not* treat the bootstrap interval’s exclusion of zero as evidence. The unbiased MMD^2 estimator is clamped at zero (Algorithm 3) and the population quantity is non-negative by construction, so a positive point estimate almost necessarily yields a CI above zero; the interval is reported to convey the *precision* of the estimate ([0.0705, 0.0801], a width of 0.0096), not as a test against the null. For the same reason we relegate the null-standardized separation Z : at $N = 500$ the observed value lies 342 null standard deviations above the null mean, but this magnitude carries no inferential content. It grows approximately linearly with N by construction, because the null variance shrinks as $O(1/N)$ while the observed statistic converges to a positive constant, and the permutation null is strongly non-Gaussian, so Z maps to no tail probability. It is reported only as a descriptive signal-to-noise ratio and should not be compared across sample sizes. Inference rests on the permutation p -value and the bootstrap interval, both of which establish separability between the real and generated distributions well beyond conventional thresholds and, unlike FID/KID/CMMD, are produced natively by the metric.

5. Discussion

Why RadioDINO and not InceptionV3 or CLIP? Backbone alignment with the target domain is not a stylistic choice; it is a prerequisite for sensitivity. InceptionV3 is trained on 1000 natural-image classes and CLIP on web image-caption pairs. Neither has been exposed to the systematic co-variation of tissue intensity, pathological contrast, and acquisition artifact that defines a radiology distribution. The clearest evidence is the CMMD rank inversion (Table 6): CLIP features place the cross-modality CT shift *closer* to real brain MRI than a weak in-domain MRI generator ($\Delta_{CT-MRI} = -0.073$, with disjoint bootstrap intervals), whereas the radiology-trained RadioDINO-s16 fidelity axis orders them correctly. The same backbone choice yields the per-image margin on the hard tier (ROC-AUC 0.819 against 0.555 for InceptionV3 and 0.582 for CLIP, both near chance).

This is not a defect of the MMD estimator but a *misalignment of the CLIP embedding space with radiology modality structure*. CLIP is trained to align images with free-text captions, so its geometry is organized around high-level, modality-agnostic semantics (“a medical scan of a brain/chest”) rather than the Hounsfield-unit, tissue-texture, and acquisition statistics that govern pixel-level radiology fidelity. Under that objective, a lung CT and a brain MRI can be nearer in caption-semantic space than a brain MRI is to a *degraded* brain MRI; the CLIP embedding is doing exactly what it was trained to do. The claim is therefore specific to this ordering task: CLIP features place a cross-modality CT shift closer to brain MRI than a weak MRI generator, where a radiology-pretrained encoder does not. It implies nothing about CLIP’s performance on natural-image evaluation or about the CMMD implementation. Whether the inversion reflects the text-alignment objective itself, or CLIP’s scale and training distribution, is not resolved by our design and would need an architecture-matched CLIP to separate.

On the a-priori layers and the discarded adaptive design The adaptive design that preceded the canonical metric selected layers by CKA pruning and a discriminability criterion. It was abandoned because selecting a layer because it maximizes measured separability is selection on the test outcome, which inflates apparent performance. The canonical metric now fixes L12 and L9, and the excluded L4 diagnostic, by representational position alone. The CKA ablation (Section 4.9) is retained only to show that this a-priori choice coincides with what an adaptive procedure would have picked for the fidelity axis (L12 is retained at every threshold): fixing the layer in advance costs no discriminability and removes the route to outcome-driven selection.

Measured limitations Four limitations remain. First, *sample efficiency*: FID is more stable than M3 for $N < 100$ (Table 11); the unbiased MMD² estimator pays for its lack of a Gaussian assumption with higher small-sample variance, so M3 should be reported at $N \geq 100$. Second, the *coverage axis* is not yet a validated claim: under mode-drop it behaves non-monotonically because k -NN radii inflate as the generated set shrinks (Section 4.4). Third, the *lesion-specificity* of M3 is real but narrow: with real expert masks it vanishes under gross erasure and appears only under subtle degradation (1.4–1.8× vs. an Inception baseline’s ~1.05×; Section 4.7). It is therefore artifact *specificity* to subtle, semantically meaningful change, not a raw-sensitivity advantage. A study with a lesion-conditioned generator is required to extend this from a metric-sensitivity test to a generator-quality claim. Fourth, the *memorization and coverage layer priors are unconfirmed*: the depth ablation of Section 4.6 shows that L9 is indistinguishable from every other block under the protocol on which the memorization axis was validated, and that no depth at all gives the coverage axis the correct sign. Only the fidelity layer L12 is independently corroborated. Fixing depths in advance still removes a route to outcome-driven tuning, but that is a statement about method discipline, not evidence that these depths are correct, and for L9 the evidence currently points the other way.

5.1. Limitations

We separate limitations that bound the present claims from questions this paper deliberately leaves open.

FID is more repeatable at any fixed N below 500 The unbiased MMD² estimator carries higher variance than FID’s Gaussian-moment estimator at every sample size we tested short of 500 (Table 11: CV 6.7% against 3.6% at $N = 100$), and can return small negative values for $N < 50$. For a study confined to a small evaluation set that only ever compares within itself, FID is the better instrument and we recommend it there. Our guidance is M3 at $N \geq 100$, and the canonical configuration uses $N = 500$. This trades against comparability rather than dominating it: the same table shows FID’s expected value moving by 2.52× across the sizes at which it is the more repeatable choice.

A design failure of our own An earlier per-image analysis correlated each candidate distance against an Isolation Forest anomaly score. The detector must be fitted in some feature space, and that space is one of those under comparison, which advantages it by construction. Running the design symmetrically (Table 4) shows each distance tracking the detector fitted in its own space four to five times more strongly than the other, in both directions. We withdraw the comparison and report the cross-tabulation instead, because the flaw is not visible without it.

We do not claim the a-priori depths are optimal The three depths are structural priors fixed by representational position (Section 3.3). This buys freedom from selection-on-outcome bias and nothing more. Our own validations are only partially informative about optimality: the near-duplicate protocol saturates at every depth, so it cannot rank them; the mode-drop check is depth-independent in direction; and only L12 has independent corroboration, from a correlational CKA analysis run on the same encoder. Establishing optimality would require measuring every depth

against every task on a reference whose composition is itself controlled, and treating the resulting profile rather than any single named block as the object of interest. We regard that as an open question and are pursuing it separately.

The permutation test answers a weak question The fidelity axis admits a permutation test where FID structurally cannot, and we report it. But its null is exact equality of the real and generated distributions, which is false by construction for any generator worth evaluating. Consistent with this, every generated row in Table 2 is rejected at the resolution floor, including a different imaging modality; the test separates none of them from each other. Its value here is confined to the null row, where it correctly declines to reject. A practitioner comparing two generators needs a null describing how far apart two *real* patient cohorts sit, which the permutation test does not provide. Constructing that null is beyond this paper’s scope.

The reference protocol is necessary but we do not quantify it Section 3.2 adopts subject-level reference construction, and Table 2 shows the null row behaving correctly under it. We have not measured how much a reported value changes as reference composition varies, how the effective sample size of a slice reference relates to its subject count, or how far the resulting distortion can travel. These are consequential questions (a reference is a property of every distributional metric, not only ours), and they are the subject of a companion study.

Lesion specificity is measured under synthetic perturbation The lesion result uses real expert glioma masks but a synthetic degradation (additive noise) confined to them. It establishes that the embedding weights tumour tissue differently from area- and texture-matched healthy tissue. It does not establish sensitivity to the failure modes that matter clinically (a hallucinated lesion, a missing one, a mis-rendered margin), which would require a lesion-conditioned generator evaluated against the same masks. The effect is also individually significant only in the upper part of the noise sweep.

Domain and scale M3 is scoped to the radiology (CT/MRI) domain in which RadioDINO was trained; our cross-modality ordering shows values are not comparable across the radiology boundary (Section 4.3). Validation is primarily on brain MRI, at $N = 500$, on one public collection. Larger-scale ($N \geq 1000$), multi-site, and chest X-ray or CT validation with ground-truth pathology masks remain future work. The metric also requires a forward pass through a 12-layer ViT-S/16: on one consumer GPU the fidelity MMD² itself takes ≈ 8 s for 500 images, and ≈ 1 –2 minutes including feature extraction.

Connection to existing metrics The three M3 axes are complementary to manifold-based metrics such as Alpha-Precision and Beta-Recall. The fidelity axis measures *feature-level* shift with a domain-calibrated kernel and a native significance test; the memorization axis targets near-duplicate leakage; the coverage axis (with the caveats above) targets diversity. We recommend reporting M3 alongside FID rather than in place of it: FID remains the more sample-stable scalar at small N , while M3 adds radiology-domain discrimination, the CMMD-style rank inversion it avoids, and per-axis diagnostics.

6. Conclusion

We introduced the M3-Score, a radiology-domain evaluation framework for generative image models built on the frozen RadioDINO-s16 ViT. Rather than a single tuned scalar, M3 reports three axes read from blocks fixed a priori by representational position: fidelity (unbiased multi-bandwidth RBF MMD² at L12), memorization (nearest-neighbour distances at L9), and coverage (real-radius manifold coverage at L4). Every reference set is drawn round-robin across subjects, because slices from one patient are near-duplicates and a reference of N slices is otherwise not a sample of N independent images.

On a subject-diverse BraTS reference the three axes order five comparison sets of increasing severity exactly, and a held-out real probe correctly registers no shift ($\widehat{\text{MMD}}^2 = 0.000$, permutation $z = -0.6$). The axes are not redundant: an unconditional DDPM attains a modest fidelity distance (0.073, bootstrap 95% CI [0.071, 0.080]) while covering only 38% of the real manifold (individually plausible images, too few distinct ones), a combination no single scalar can express.

The coverage axis carries a methodological result of independent interest. Specified as k -NN manifold recall it fails its own mode-drop check at every one of the twelve encoder blocks, in the wrong direction, because that estimator fits its hyperspheres on the set being subsampled. Specified with real-set radii it falls monotonically at all twelve. We

report both, since a sanity check that nothing fails demonstrates nothing. The caution generalizes: any protocol that subsamples one of the two sets being compared can invert a k -NN precision/recall statistic while appearing to measure diversity.

Three results characterize the backbone and the estimator. With the scoring rule held fixed and only the feature space varied, RadioDINO separates real brain MRI from in-domain DDPM output at ROC-AUC 0.819, against 0.555 for the InceptionV3 space underlying FID and 0.582 for the CLIP space underlying CMMD, both near chance on the one tier where the spaces differ at all. CMMD rank-inverts a cross-modality CT shift against a weak same-modality generator, where M3 and FID order both correctly. And on identical data across a twentyfold change in evaluation size, the M3 value drifts by $1.05\times$ while FID drifts by $2.52\times$: two studies quoting FID against different-sized references are not comparing the same quantity, and the gap between them can exceed the gap between a good and a poor generator. With real expert glioma masks and area- and texture-matched controls, M3 is $1.4\text{--}1.8\times$ more responsive to tumour-region than to control-region perturbation, where an Inception-MMD baseline is not.

We are explicit about what remains open. FID is the more repeatable statistic at any fixed N below 500, and we recommend it over M3 for studies confined to small evaluation sets that only ever compare within themselves. We also report a design failure of our own: an earlier per-image analysis correlated each distance against an Isolation Forest fitted in one of the feature spaces under test, which advantages that space by construction; running it symmetrically shows the diagonal dominating by four to five times in both directions, so it ranks nothing and we withdraw the comparison. The a-priori depths are unbiased by construction but not shown to be optimal. The permutation test answers a weak question that every generator rejects, and the null a practitioner actually needs (how far apart two real patient cohorts sit) is not one this paper constructs. And while we adopt subject-level reference construction and show it makes the null case behave, we do not quantify how much reference composition can move a reported value. Those last two questions are the subject of a companion study; the open-source implementation, the a-priori configuration, and the acceptance criteria for every validation run are released with this paper.

A. Legacy Adaptive Aggregation and CKA Layer Selection

This appendix specifies the one component of the superseded adaptive design that the reported results still depend on: the CKA-based layer selection used for the post-hoc check of Section 4.9, which is what shows L12 to be the one layer retained at every redundancy threshold. It is *not* part of the canonical M3-Score (Algorithm 1, Section 3.3): the canonical axes read fixed layers and require none of this machinery, so no layer, threshold, or weight is selected on test outcomes.

The remainder of the abandoned design (the adaptive aggregation loop and the entropy-stability-uniqueness layer weighting) is not reproduced here, since no result in this paper uses it. Its full specification and reference implementation are available in the released code repository (Section 6).

A.1. Post-Hoc Validation: CKA-Based Layer Analysis

Motivation Adjacent transformer layers in DINOv2-style encoders frequently learn representations of high mutual similarity, particularly in the early and final blocks (Kornblith et al., 2019). Including such layers would introduce correlated and computationally redundant contributions to the final score. The CKA (Kornblith et al., 2019) is used to identify a non-redundant subset $\mathcal{L}$ of the available layers L .

Linear CKA Given two embedding matrices $E^{(\ell_1)}, E^{(\ell_2)} \in \mathbb{R}^{N_r \times D}$ extracted from a reference set of N_r real images, linear CKA is

$$\text{CKA}(E^{(\ell_1)}, E^{(\ell_2)}) = \frac{\|\tilde{E}^{(\ell_1)\top} \tilde{E}^{(\ell_2)}\|_F^2}{\|\tilde{E}^{(\ell_1)\top} \tilde{E}^{(\ell_1)}\|_F^2 \cdot \|\tilde{E}^{(\ell_2)\top} \tilde{E}^{(\ell_2)}\|_F^2}^{1/2}, \quad (12)$$

where $\tilde{E}^{(\ell)} = E^{(\ell)} - \mathbf{1}\bar{e}^{(\ell)\top}$ is the column-centered matrix, $\bar{e}^{(\ell)} = \frac{1}{N_r} \sum_i E_{i,:}^{(\ell)}$ is the column mean, and $\|\cdot\|_F$ denotes the Frobenius norm. $\text{CKA}(E^{(\ell_1)}, E^{(\ell_2)}) \in [0, 1]$, with value 1 indicating identical representational geometry.

Evaluating Eq. (12) directly requires forming $(D \times D)$ covariance matrices, whose denominator scales as $\|\cdot\|_F^4 \sim D^4$ and is numerically unstable. Following the HSIC reformulation in (Kornblith et al., 2019), we instead compute CKA via $(N_r \times N_r)$ gram matrices, which are independent of feature dimension:

Algorithm 5 CKA-based greedy layer selection (CKAPRUNE)**Require:** Encoder Φ , reference set $\mathcal{X}_r^{(N_r)}$, threshold τ **Ensure:** Active layer set $\mathcal{L} \subseteq \{1, \dots, L\}$

```

1: for  $\ell = 1, \dots, L$  do
2:    $E^{(\ell)} \leftarrow \text{EXTRACTFEATURES}(\Phi, \mathcal{X}_r^{(N_r)}, \{\ell\})$ 
3:    $\tilde{E}^{(\ell)} \leftarrow E^{(\ell)} - \mathbf{1}\bar{e}^{(\ell)\top}$  ▷ column-center, Eq. (12)
4:    $K^{(\ell)} \leftarrow \tilde{E}^{(\ell)} \tilde{E}^{(\ell)\top}$  ▷  $(N_r \times N_r)$  gram matrix, Eq. (13)
5: end for
6:  $\mathcal{L} \leftarrow \{L\}$  ▷ layer  $L$  is retained unconditionally
7: for  $\ell = L - 1, L - 2, \dots, 1$  do
8:    $\text{redundant} \leftarrow \text{false}$ 
9:   for each  $\ell' \in \mathcal{L}$  do
10:     $\rho \leftarrow \langle K^{(\ell)}, K^{(\ell')} \rangle_F / (\|K^{(\ell)}\|_F \cdot \|K^{(\ell')}\|_F)$  ▷ Eq. (14)
11:    if  $\rho > \tau$  then
12:       $\text{redundant} \leftarrow \text{true}$ 
13:      break
14:    end if
15:   end for
16:   if  $\neg \text{redundant}$  then
17:      $\mathcal{L} \leftarrow \mathcal{L} \cup \{\ell\}$ 
18:   end if
19: end for
20: return  $\mathcal{L}$ 

```

$$K^{(\ell)} = \tilde{E}^{(\ell)} \tilde{E}^{(\ell)\top} \in \mathbb{R}^{N_r \times N_r}, \quad (13)$$

$$\text{CKA}(\ell_1, \ell_2) = \frac{\langle K^{(\ell_1)}, K^{(\ell_2)} \rangle_F}{\|K^{(\ell_1)}\|_F \cdot \|K^{(\ell_2)}\|_F}, \quad (14)$$

where $\langle A, B \rangle_F = \sum_{ij} A_{ij} B_{ij}$ is the Frobenius inner product, proportional to the Hilbert-Schmidt Independence Criterion.

Greedy backward selection Given $\tau \in (0, 1)$ (default $\tau = 0.80$), a greedy pass iterating from layer $L - 1$ to layer 1 adds a layer to $\mathcal{L}$ only if its CKA similarity to every layer already in $\mathcal{L}$ is below τ . Layer L is always retained as the anchor. The full procedure is stated as Algorithm 5.

On BraTS brain MRI with $\tau = 0.80$, the procedure selects $M = 3$ layers $\mathcal{L} = \{1, 3, 12\}$, spanning early, mid, and deep transformer blocks. The number M of retained layers is data-adaptive and varies with both the imaging modality and the CKA threshold τ ; it is reported alongside S for reproducibility. In the BraTS evaluation, layers 1, 3, and 12 are retained because their pairwise CKA similarities all fall below $\tau = 0.80$, indicating that each captures a distinct aspect of the radiology-specific feature hierarchy (Dosovitskiy et al., 2021; Oquab et al., 2023): early layers encode local texture and edge responses, mid-network layers encode tissue boundaries and coarse anatomy, and the final layer encodes semantic pathology structure. CKA layer selection is performed once per evaluation session on $N_r = 20$ images using CLS token features and the resulting $\mathcal{L}$ is reused for the full evaluation pass.

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